

# Top 100 SAS Variant-Disease Hits

Ranked by unmet-need index, L2G score, and SAS global MAF. Generated from  
ot\_sas\_gi\_vep115\_plof\_deleterious\_missense\_disease\_gene\_variant\_map.india\_exclusive.score\_ge\_0.15.parquet.

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48. 48. FMN2 - cardiac arrest L2G 0.69; unmet 5.03
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53. 53. TP63 - lung adenocarcinoma L2G 0.906; unmet 5
54. 54. SECISBP2L - lung adenocarcinoma L2G 0.884; unmet 5
55. 55. HNF1B - lung adenocarcinoma L2G 0.865; unmet 5
56. 56. FOXP4 - lung adenocarcinoma L2G 0.575; unmet 5
57. 57. TCF7L2 - lung adenocarcinoma L2G 0.435; unmet 5
58. 58. RTKL1 - lung adenocarcinoma L2G 0.277; unmet 5
59. 59. CYP2A6 - lung adenocarcinoma L2G 0.238; unmet 5
60. 60. GPC5 - gastric cancer L2G 0.869; unmet 5
61. 61. POU5F1B - gastric cancer L2G 0.828; unmet 5
62. 62. HNF1B - gastric cancer L2G 0.71; unmet 5
63. 63. STPG2 - gastric cancer L2G 0.342; unmet 5
64. 64. NR2E3 - gastric cancer L2G 0.252; unmet 5
65. 65. L1TD1 - neuroblastoma L2G 0.931; unmet 4.96
66. 66. KANSL1L - neuroblastoma L2G 0.692; unmet 4.96
67. 67. WRN - pneumococcal pneumonia L2G 0.585; unmet 4.94
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88. 88. ANKS1B - amyotrophic lateral sclerosis L2G 0.75; unmet 4.81
89. 89. ACOXL - amyotrophic lateral sclerosis L2G 0.674; unmet 4.81
90. 90. G2E3 - amyotrophic lateral sclerosis L2G 0.643; unmet 4.81
91. 91. CRYZ - amyotrophic lateral sclerosis L2G 0.608; unmet 4.81
92. 92. NFIA - amyotrophic lateral sclerosis L2G 0.52; unmet 4.81
93. 93. WDR72 - amyotrophic lateral sclerosis L2G 0.459; unmet 4.81
94. 94. NDST4 - amyotrophic lateral sclerosis L2G 0.451; unmet 4.81
95. 95. CCNA1 - amyotrophic lateral sclerosis L2G 0.367; unmet 4.81
96. 96. KIAA0232 - amyotrophic lateral sclerosis L2G 0.311; unmet 4.81
97. 97. MTNR1B - amyotrophic lateral sclerosis L2G 0.305; unmet 4.81
98. 98. GGT5 - amyotrophic lateral sclerosis L2G 0.169; unmet 4.81
99. 99. POU5F1B - hepatocellular carcinoma L2G 0.828; unmet 4.79
100. 100. HNF1B - hepatocellular carcinoma L2G 0.71; unmet 4.79

# 1. PLCE1 - thoracic aortic aneurysm

ENSG00000138193 | EFO\_0004282

L2G score  
**0.851**

Unmet need index  
**5.73**

SAS global MAF  
**0.3109**

Genome India global MAF  
**0.3439**

Variants  
**2**

Interpretations  
**Likely consequential**

## Disease Description

An aneurysm formed in the wall of the proximal portion of the descending aorta proceeding from the arch of the aorta.

## Gene Summary

| phospholipase C epsilon 1

## Known Drugs / Clinical Investigation

| No same target-disease clinical indication found in the downloaded Open Targets drug tables.

## Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.851 across 2 evidence rows and 2 study loci. No datasource-level direct association summary was found in the downloaded association table.

## SAS / Indian Prevalence

Rare

## Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|----------|-----------------|--------|--------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 10_94032003_G_C | missense_variant | Likely consequential    | 0.001762 | 0.001842        | 0      | 0      | 21.2       | 0.6314           |           | 0.24  | 0.2131   | 0.3109            | 0.3439                     |
| 10_94171330_G_T | missense_variant | Likely consequential    | 0.3097   | 0.3427          | 0.1691 | 0.1296 | 23.8       | 0.3728           | 0.5887    | 0.086 | 0.01078  | 0.3109            | 0.3439                     |

2. ELN - thoracic aortic aneurysm

ENSG00000049540 | EFO\_0004282

|                                     |                          |                                         |
|-------------------------------------|--------------------------|-----------------------------------------|
| L2G score<br>0.801                  | Unmet need index<br>5.73 | SAS global MAF<br>0.0003826             |
| Genome India global MAF<br>0.002047 | Variants<br>1            | Interpretations<br>Likely consequential |

Disease Description

An aneurysm formed in the wall of the proximal portion of the descending aorta proceeding from the arch of the aorta.

Gene Summary

| elastin

Known Drugs / Clinical Investigation

| No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.801 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid         | consequence      | missense_interpretation | AF_sas    | genome_india_af | AF_nfe    | AF_fin  | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|---------------|------------------|-------------------------|-----------|-----------------|-----------|---------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 7_74042640_TC | missense_variant | Likely consequential    | 0.0003826 | 0.002047        | 0.0002662 | 0.00732 | 20.7       | 0.834            |           | 0.125 | 0.1057   | 0.0003826         | 0.002047                   |

### 3. ULK4 - thoracic aortic aneurysm

ENSG00000168038 | EFO\_0004282

|                                     |                                 |                                        |
|-------------------------------------|---------------------------------|----------------------------------------|
| L2G score<br><b>0.774</b>           | Unmet need index<br><b>5.73</b> | SAS global MAF<br><b>0.007181</b>      |
| Genome India global MAF<br><b>0</b> | Variants<br><b>2</b>            | Interpretations<br><b>not_missense</b> |

#### Disease Description

An aneurysm formed in the wall of the proximal portion of the descending aorta proceeding from the arch of the aorta.

#### Gene Summary

unc-51 like kinase 4

#### Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

#### Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.774 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

#### SAS / Indian Prevalence

Rare

#### Alleles and Scores

| varid           | consequence        | missense_interpretation | AF_sas   | genome_india_af | AF_nfe    | AF_fin   | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|--------------------|-------------------------|----------|-----------------|-----------|----------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 3_41819492_CT_C | frameshift_variant | not_missense            | 0.00245  |                 | 0.0002843 | 0.001709 | 28.5       |                  |           |       |          | 0.007181          | 0                          |
| 3_41819492_C_CT | frameshift_variant | not_missense            | 0.004742 |                 | 0.001492  | 0.003278 | 33         |                  |           |       |          | 0.007181          | 0                          |

4. ABO - pancreatic adenocarcinoma

ENSG00000175164 | EFO\_1000044

|                                    |                          |                                 |
|------------------------------------|--------------------------|---------------------------------|
| L2G score<br>0.836                 | Unmet need index<br>5.35 | SAS global MAF<br>0.04705       |
| Genome India global MAF<br>0.03973 | Variants<br>2            | Interpretations<br>not_missense |

Disease Description

A carcinoma that arises from glandular epithelial cells of the pancreas

Gene Summary

ABO, alpha 1-3-N-acetylgalactosaminyltransferase and alpha 1-3-galactosyltransferase

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.836 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid           | consequence        | missense_interpretation | AF_sas  | genome_india_af | AF_nfe   | AF_fin    | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|--------------------|-------------------------|---------|-----------------|----------|-----------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 9_133256189_C_T | stop_gained        | not_missense            | 0.0329  | 0.02446         | 0.01796  | 0.01162   | 51         |                  |           |       |          | 0.04705           | 0.03973                    |
| 9_133256234_G_T | frameshift_variant | not_missense            | 0.01463 | 0.01566         | 0.002961 | 0.0003027 | 17.65      |                  |           |       |          | 0.04705           | 0.03973                    |

5. HNF1B - squamous cell lung carcinoma

ENSG00000275410 | EFO\_0000708

|                                     |                                 |                                        |
|-------------------------------------|---------------------------------|----------------------------------------|
| L2G score<br><b>0.865</b>           | Unmet need index<br><b>5.31</b> | SAS global MAF<br><b>0.0169</b>        |
| Genome India global MAF<br><b>0</b> | Variants<br><b>1</b>            | Interpretations<br><b>not_missense</b> |

Disease Description

A carcinoma arising from squamous bronchial epithelial cells. It may be keratinizing or non-keratinizing. Keratinizing squamous cell carcinoma is characterized by the presence of keratinization, pearl formation, and/or intercellular bridges. Non-keratinizing squamous cell carcinoma is characterized by the absence of keratinization, pearl formation, and intercellular bridges. Cigarette smoking and arsenic exposure are strongly associated with squamous cell lung carcinoma.

Gene Summary

HNF1 homeobox B

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.865 across 2 evidence rows and 2 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid                  | consequence        | missense_interpretation | AF_sas | genome_india_af | AF_nfe | AF_fin  | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|------------------------|--------------------|-------------------------|--------|-----------------|--------|---------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 17_37710582_GTCACCAT_G | frameshift_variant | not_missense            | 0.0169 |                 | 0.0023 | 0.02255 | 35         |                  |           |       |          | 0.0169            | 0                          |



6. CYP2A6 - squamous cell lung carcinoma

ENSG00000255974 | EFO\_0000708

L2G score  
0.238

Unmet need index  
5.31

SAS global MAF  
0.01779

Genome India global MAF  
0.00764

Variants  
3

Interpretations  
Likely consequential;  
not\_missense

Disease Description

A carcinoma arising from squamous bronchial epithelial cells. It may be keratinizing or non-keratinizing. Keratinizing squamous cell carcinoma is characterized by the presence of keratinization, pearl formation, and/or intercellular bridges. Non-keratinizing squamous cell carcinoma is characterized by the absence of keratinization, pearl formation, and intercellular bridges. Cigarette smoking and arsenic exposure are strongly associated with squamous cell lung carcinoma.

Gene Summary

cytochrome P450 family 2 subfamily A member 6

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.238 across 3 evidence rows and 3 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe   | AF_fin   | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|----------|-----------------|----------|----------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 19_40843845_C_A | missense_variant | Likely consequential    | 0.001052 |                 | 0.000346 | 0.001268 | 23.1       | 0.6058            |            | 0.147 | 0.1383   | 0.01779           | 0.00764                    |
| 19_40843881_G_T | stop_gained      | not_missense            | 0.009121 |                 | 0.01919  | 0.003588 | 33         |                   |            |       |          | 0.01779           | 0.00764                    |
| 19_40848304_C_A | missense_variant | Likely consequential    | 0.007708 | 0.00764         | 0        | 0        | 29         | 0.7887            |            | 0.312 | 0.2805   | 0.01779           | 0.00764                    |

7. ERC1 - squamous cell lung carcinoma

ENSG00000082805 | EFO\_0000708

|                              |                          |                                         |
|------------------------------|--------------------------|-----------------------------------------|
| L2G score<br>0.187           | Unmet need index<br>5.31 | SAS global MAF<br>0.001693              |
| Genome India global MAF<br>0 | Variants<br>1            | Interpretations<br>Likely consequential |

Disease Description

A carcinoma arising from squamous bronchial epithelial cells. It may be keratinizing or non-keratinizing. Keratinizing squamous cell carcinoma is characterized by the presence of keratinization, pearl formation, and/or intercellular bridges. Non-keratinizing squamous cell carcinoma is characterized by the absence of keratinization, pearl formation, and intercellular bridges. Cigarette smoking and arsenic exposure are strongly associated with squamous cell lung carcinoma.

Gene Summary

ELKS/RAB6-interacting/CAST family member 1

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.187 across 4 evidence rows and 4 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid         | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|---------------|------------------|-------------------------|----------|-----------------|--------|--------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 12_1028211_GC | missense_variant | Likely consequential    | 0.001693 |                 | 0      | 0      | 31         | 0.9299           |           | 0.779 | 0.1305   | 0.001693          | 0                          |

## 8. ABO - Ischemic stroke

ENSG00000175164 | HP\_0002140

|                                           |                                 |                                        |
|-------------------------------------------|---------------------------------|----------------------------------------|
| L2G score<br><b>0.961</b>                 | Unmet need index<br><b>5.27</b> | SAS global MAF<br><b>0.04705</b>       |
| Genome India global MAF<br><b>0.03973</b> | Variants<br><b>2</b>            | Interpretations<br><b>not_missense</b> |

### Disease Description

Acute ischemic stroke (AIS) is defined by the sudden loss of blood flow to an area of the brain with the resulting loss of neurologic function. It is caused by thrombosis or embolism that occludes a cerebral vessel supplying a specific area of the brain. During a vessel occlusion, there is a core area where damage to the brain is irreversible and an area of penumbra where the brain has lost function owing to decreased blood flow but is not irreversibly injured.

### Gene Summary

ABO, alpha 1-3-N-acetylgalactosaminyltransferase and alpha 1-3-galactosyltransferase

### Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

### Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.961 across 8 evidence rows and 8 study loci. No datasource-level direct association summary was found in the downloaded association table.

### SAS / Indian Prevalence

Common

### Alleles and Scores

| varid           | consequence        | missense_interpretation | AF_sas  | genome_india_af | AF_nfe   | AF_fin    | cadd_phred | am_pathogenicity | EVE_S CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|--------------------|-------------------------|---------|-----------------|----------|-----------|------------|------------------|------------|-------|----------|-------------------|----------------------------|
| 9_133256189_C_T | stop_gained        | not_missense            | 0.0329  | 0.02446         | 0.01796  | 0.01162   | 51         |                  |            |       |          | 0.04705           | 0.03973                    |
| 9_133256234_G_T | frameshift_variant | not_missense            | 0.01463 | 0.01566         | 0.002961 | 0.0003027 | 17.65      |                  |            |       |          | 0.04705           | 0.03973                    |

9. HDLBP - Ischemic stroke

ENSG00000115677 | HP\_0002140

L2G score  
**0.763**

Unmet need index  
**5.27**

SAS global MAF  
**0.04532**

Genome India global MAF  
**0**

Variants  
**1**

Interpretations  
**Likely consequential**

Disease Description

Acute ischemic stroke (AIS) is defined by the sudden loss of blood flow to an area of the brain with the resulting loss of neurologic function. It is caused by thrombosis or embolism that occludes a cerebral vessel supplying a specific area of the brain. During a vessel occlusion, there is a core area where damage to the brain is irreversible and an area of penumbra where the brain has lost function owing to decreased blood flow but is not irreversibly injured.

Gene Summary

high density lipoprotein binding protein

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.763 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Common

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas  | genome_india_af | AF_nfe   | AF_fin  | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|---------|-----------------|----------|---------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 2_241239714_A_C | missense_variant | Likely consequential    | 0.04532 |                 | 0.006525 | 0.03564 | 24.6       | 0.9741           |           | 0.575 | 0.05503  | 0.04532           | 0                          |

## 10. CUX2 - Ischemic stroke

ENSG00000111249 | HP\_0002140

|                                     |                                 |                                                                  |
|-------------------------------------|---------------------------------|------------------------------------------------------------------|
| L2G score<br><b>0.717</b>           | Unmet need index<br><b>5.27</b> | SAS global MAF<br><b>0.006762</b>                                |
| Genome India global MAF<br><b>0</b> | Variants<br><b>2</b>            | Interpretations<br><b>Likely consequential;<br/>not_missense</b> |

### Disease Description

Acute ischemic stroke (AIS) is defined by the sudden loss of blood flow to an area of the brain with the resulting loss of neurologic function. It is caused by thrombosis or embolism that occludes a cerebral vessel supplying a specific area of the brain. During a vessel occlusion, there is a core area where damage to the brain is irreversible and an area of penumbra where the brain has lost function owing to decreased blood flow but is not irreversibly injured.

### Gene Summary

cut like homeobox 2

### Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

### Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.717 across 9 evidence rows and 9 study loci. No datasource-level direct association summary was found in the downloaded association table.

### SAS / Indian Prevalence

Common

### Alleles and Scores

| varid              | consequence        | missense_interpretation | AF_sas   | genome_india_af | AF_nfe   | AF_fin   | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|--------------------|--------------------|-------------------------|----------|-----------------|----------|----------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 12_111347618_T_TGC | frameshift_variant | not_missense            | 0.002081 |                 | 0.007879 | 3.78e-05 | 32         |                   |            |       |          | 0.006762          | 0                          |
| 12_111347622_A_C   | missense_variant   | Likely consequential    | 0.00469  |                 | 0.02853  | 0.00159  | 21.5       | 0.0743            |            | 0.072 | 0.737    | 0.006762          | 0                          |

# 11. SPPL2A - Ischemic stroke

ENSG00000138600 | HP\_0002140

L2G score  
**0.45**

Unmet need index  
**5.27**

SAS global MAF  
**0.001057**

Genome India global MAF  
**0.0008187**

Variants  
**1**

Interpretations  
**Likely consequential**

## Disease Description

Acute ischemic stroke (AIS) is defined by the sudden loss of blood flow to an area of the brain with the resulting loss of neurologic function. It is caused by thrombosis or embolism that occludes a cerebral vessel supplying a specific area of the brain. During a vessel occlusion, there is a core area where damage to the brain is irreversible and an area of penumbra where the brain has lost function owing to decreased blood flow but is not irreversibly injured.

## Gene Summary

signal peptide peptidase like 2A

## Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

## Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.45 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

## SAS / Indian Prevalence

Common

## Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe   | AF_fin    | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|----------|-----------------|----------|-----------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 15_50722170_A_T | missense_variant | Likely consequential    | 0.001057 | 0.0008187       | 0.002762 | 0.0001206 | 24.3       | 0.9626           |           | 0.538 | 0.08175  | 0.001057          | 0.0008187                  |

## 12. ADH5 - Ischemic stroke

ENSG00000197894 | HP\_0002140

|                                            |                                 |                                                |
|--------------------------------------------|---------------------------------|------------------------------------------------|
| L2G score<br><b>0.334</b>                  | Unmet need index<br><b>5.27</b> | SAS global MAF<br><b>0.0001752</b>             |
| Genome India global MAF<br><b>0.001177</b> | Variants<br><b>1</b>            | Interpretations<br><b>Likely consequential</b> |

### Disease Description

Acute ischemic stroke (AIS) is defined by the sudden loss of blood flow to an area of the brain with the resulting loss of neurologic function. It is caused by thrombosis or embolism that occludes a cerebral vessel supplying a specific area of the brain. During a vessel occlusion, there is a core area where damage to the brain is irreversible and an area of penumbra where the brain has lost function owing to decreased blood flow but is not irreversibly injured.

### Gene Summary

alcohol dehydrogenase 5 (class III), chi polypeptide

### Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

### Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.334 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

### SAS / Indian Prevalence

Common

### Alleles and Scores

| varid          | consequence      | missense_interpretation | AF_sas    | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|----------------|------------------|-------------------------|-----------|-----------------|--------|--------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 4_99072585_T_G | missense_variant | Likely consequential    | 0.0001752 | 0.001177        | 0      | 0      | 22.6       | 0.7039           |           | 0.222 | 0.4195   | 0.0001752         | 0.001177                   |

### 13. MMP3 - Ischemic stroke

ENSG00000149968 | HP\_0002140

L2G score  
**0.277**

Unmet need index  
**5.27**

SAS global MAF  
**0.09613**

Genome India global MAF  
**0**

Variants  
**2**

Interpretations  
**not\_missense**

#### Disease Description

Acute ischemic stroke (AIS) is defined by the sudden loss of blood flow to an area of the brain with the resulting loss of neurologic function. It is caused by thrombosis or embolism that occludes a cerebral vessel supplying a specific area of the brain. During a vessel occlusion, there is a core area where damage to the brain is irreversible and an area of penumbra where the brain has lost function owing to decreased blood flow but is not irreversibly injured.

#### Gene Summary

matrix metallopeptidase 3

#### Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

#### Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.277 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

#### SAS / Indian Prevalence

Common

#### Alleles and Scores

| varid            | consequence          | missense_interpretation | AF_sas  | genome_india_af | AF_nfe   | AF_fin  | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|------------------|----------------------|-------------------------|---------|-----------------|----------|---------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 11_102842429_A_T | splice_donor_variant | not_missense            | 0.06631 |                 | 0.008394 | 0.07395 | 27.2       |                  |           |       |          | 0.09613           | 0                          |
| 11_102842430_C_T | splice_donor_variant | not_missense            | 0.03193 |                 | 0.008242 | 0.05908 | 29         |                  |           |       |          | 0.09613           | 0                          |



14. ICA1L - Ischemic stroke

ENSG00000163596 | HP\_0002140

|                                   |                          |                                 |
|-----------------------------------|--------------------------|---------------------------------|
| L2G score<br>0.231                | Unmet need index<br>5.27 | SAS global MAF<br>1.16e-05      |
| Genome India global MAF<br>0.9999 | Variants<br>1            | Interpretations<br>not_missense |

Disease Description

Acute ischemic stroke (AIS) is defined by the sudden loss of blood flow to an area of the brain with the resulting loss of neurologic function. It is caused by thrombosis or embolism that occludes a cerebral vessel supplying a specific area of the brain. During a vessel occlusion, there is a core area where damage to the brain is irreversible and an area of penumbra where the brain has lost function owing to decreased blood flow but is not irreversibly injured.

Gene Summary

islet cell autoantigen 1 like

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.231 across 2 evidence rows and 2 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Common

Alleles and Scores

| varid            | consequence        | missense_interpretation | AF_sas   | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|------------------|--------------------|-------------------------|----------|-----------------|--------|--------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 2_202788834_G_GA | frameshift_variant | not_missense            | 1.16e-05 | 0.9999          | 0      | 0      |            |                  |           |       |          | 1.16e-05          | 0.9999                     |

15. PLCE1 - esophageal squamous cell carcinoma

ENSG00000138193 | EFO\_0005922

L2G score  
0.294

Unmet need index  
5.25

SAS global MAF  
0.3109

Genome India global MAF  
0.3439

Variants  
2

Interpretations  
Likely consequential

Disease Description

Esophageal squamous cell carcinoma (ESCC) is a type of esophageal carcinoma (EC) that can affect any part of the esophagus, but is usually located in the upper or middle third.

Gene Summary

| phospholipase C epsilon 1

Known Drugs / Clinical Investigation

| No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.294 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|----------|-----------------|--------|--------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 10_94032003_G_C | missense_variant | Likely consequential    | 0.001762 | 0.001842        | 0      | 0      | 21.2       | 0.6314           |           | 0.24  | 0.2131   | 0.3109            | 0.3439                     |
| 10_94171330_G_T | missense_variant | Likely consequential    | 0.3097   | 0.3427          | 0.1691 | 0.1296 | 23.8       | 0.3728           | 0.5887    | 0.086 | 0.01078  | 0.3109            | 0.3439                     |

16. WDR72 - chronic kidney disease

ENSG00000166415 | EFO\_0003884

L2G score  
**0.963**

Unmet need index  
**5.24**

SAS global MAF  
**0.2429**

Genome India global MAF  
**0.259**

Variants  
**1**

Interpretations  
**Likely consequential**

Disease Description

The end-stage of CHRONIC RENAL INSUFFICIENCY. It is characterized by the severe irreversible kidney damage (as measured by the level of PROTEINURIA) and the reduction in GLOMERULAR FILTRATION RATE to less than 15 ml per min (Kidney Foundation: Kidney Disease Outcome Quality Initiative, 2002). These patients generally require HEMODIALYSIS or KIDNEY TRANSPLANTATION.

Gene Summary

WD repeat domain 72

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.963 across 8 evidence rows and 8 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Ultra-Orphan

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|--------|-----------------|--------|--------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 15_53615751_G_A | missense_variant | Likely consequential    | 0.2429 | 0.259           | 0.2251 | 0.257  | 25.2       | 0.3382            | 0.7526     | 0.528 | 0.02449  | 0.2429            | 0.259                      |

17. SHROOM3 - chronic kidney disease

ENSG00000138771 | EFO\_0003884

|                                     |                          |                                         |
|-------------------------------------|--------------------------|-----------------------------------------|
| L2G score<br>0.885                  | Unmet need index<br>5.24 | SAS global MAF<br>0.003458              |
| Genome India global MAF<br>0.005114 | Variants<br>2            | Interpretations<br>Likely consequential |

Disease Description

The end-stage of CHRONIC RENAL INSUFFICIENCY. It is characterized by the severe irreversible kidney damage (as measured by the level of PROTEINURIA) and the reduction in GLOMERULAR FILTRATION RATE to less than 15 ml per min (Kidney Foundation: Kidney Disease Outcome Quality Initiative, 2002). These patients generally require HEMODIALYSIS or KIDNEY TRANSPLANTATION.

Gene Summary

shroom family member 3

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.885 across 3 evidence rows and 3 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Ultra-Orphan

Alleles and Scores

| varid          | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe   | AF_fin    | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|----------------|------------------|-------------------------|----------|-----------------|----------|-----------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 4_76555619_G_T | missense_variant | Likely consequential    | 0.001218 | 0.0006652       | 0.002349 | 0.0007865 | 27.8       | 0.7298            |            | 0.506 | 0.1086   | 0.003458          | 0.005114                   |
| 4_76741205_T_C | missense_variant | Likely consequential    | 0.002243 | 0.004451        | 0        | 0         | 25.5       | 0.7435            |            | 0.493 | 0.3707   | 0.003458          | 0.005114                   |

18. PDILT - chronic kidney disease

ENSG00000169340 | EFO\_0003884

L2G score  
**0.884**

Unmet need index  
**5.24**

SAS global MAF  
**0.003536**

Genome India global MAF  
**0.002712**

Variants  
**1**

Interpretations  
**Likely consequential**

Disease Description

The end-stage of CHRONIC RENAL INSUFFICIENCY. It is characterized by the severe irreversible kidney damage (as measured by the level of PROTEINURIA) and the reduction in GLOMERULAR FILTRATION RATE to less than 15 ml per min (Kidney Foundation: Kidney Disease Outcome Quality Initiative, 2002). These patients generally require HEMODIALYSIS or KIDNEY TRANSPLANTATION.

Gene Summary

protein disulfide isomerase like, testis expressed

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.884 across 2e+01 evidence rows and 2e+01 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Ultra-Orphan

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe   | AF_fin   | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|----------|-----------------|----------|----------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 16_20362577_G_A | missense_variant | Likely consequential    | 0.003536 | 0.002712        | 0.006165 | 0.001292 | 23.5       | 0.3062           |           | 0.586 | 0.04297  | 0.003536          | 0.002712                   |

19. TCF7L2 - chronic kidney disease

ENSG00000148737 | EFO\_0003884

L2G score  
**0.884**

Unmet need index  
**5.24**

SAS global MAF  
**0.002843**

Genome India global MAF  
**0**

Variants  
**1**

Interpretations  
**Likely consequential**

Disease Description

The end-stage of CHRONIC RENAL INSUFFICIENCY. It is characterized by the severe irreversible kidney damage (as measured by the level of PROTEINURIA) and the reduction in GLOMERULAR FILTRATION RATE to less than 15 ml per min (Kidney Foundation: Kidney Disease Outcome Quality Initiative, 2002). These patients generally require HEMODIALYSIS or KIDNEY TRANSPLANTATION.

Gene Summary

transcription factor 7 like 2

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.884 across 9 evidence rows and 9 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Ultra-Orphan

Alleles and Scores

| varid            | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe   | AF_fin   | cadd_phred | am_pathogenicity | EVE_S CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|------------------|------------------|-------------------------|----------|-----------------|----------|----------|------------|------------------|------------|-------|----------|-------------------|----------------------------|
| 10_113165649_T_C | missense_variant | Likely consequential    | 0.002843 |                 | 0.001924 | 0.002653 | 26.4       | 0.0881           | 0.0369     | 0.521 | 0.03355  | 0.002843          | 0                          |

20. BCAS3 - chronic kidney disease

ENSG00000141376 | EFO\_0003884

|                                     |                          |                                         |
|-------------------------------------|--------------------------|-----------------------------------------|
| L2G score<br>0.639                  | Unmet need index<br>5.24 | SAS global MAF<br>0.000116              |
| Genome India global MAF<br>0.001177 | Variants<br>1            | Interpretations<br>Likely consequential |

Disease Description

The end-stage of CHRONIC RENAL INSUFFICIENCY. It is characterized by the severe irreversible kidney damage (as measured by the level of PROTEINURIA) and the reduction in GLOMERULAR FILTRATION RATE to less than 15 ml per min (Kidney Foundation: Kidney Disease Outcome Quality Initiative, 2002). These patients generally require HEMODIALYSIS or KIDNEY TRANSPLANTATION.

Gene Summary

BCAS3 microtubule associated cell migration factor

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.639 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Ultra-Orphan

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_S CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|----------|-----------------|--------|--------|------------|------------------|------------|-------|----------|-------------------|----------------------------|
| 17_61040883_C_T | missense_variant | Likely consequential    | 0.000116 | 0.001177        | 0      | 0      | 27         |                  |            | 0.21  | 0.6974   | 0.000116          | 0.001177                   |

21. C6orf118 - chronic kidney disease

ENSG00000112539 | EFO\_0003884

|                                    |                          |                                 |
|------------------------------------|--------------------------|---------------------------------|
| L2G score<br>0.525                 | Unmet need index<br>5.24 | SAS global MAF<br>0.001136      |
| Genome India global MAF<br>0.00133 | Variants<br>1            | Interpretations<br>not_missense |

Disease Description

The end-stage of CHRONIC RENAL INSUFFICIENCY. It is characterized by the severe irreversible kidney damage (as measured by the level of PROTEINURIA) and the reduction in GLOMERULAR FILTRATION RATE to less than 15 ml per min (Kidney Foundation: Kidney Disease Outcome Quality Initiative, 2002). These patients generally require HEMODIALYSIS or KIDNEY TRANSPLANTATION.

Gene Summary

chromosome 6 open reading frame 118

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.525 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Ultra-Orphan

Alleles and Scores

| varid           | consequence | missense_interpretation | AF_sas   | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_S CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|-------------|-------------------------|----------|-----------------|--------|--------|------------|------------------|------------|-------|----------|-------------------|----------------------------|
| 6_165309577_C_A | stop_gained | not_missense            | 0.001136 | 0.00133         | 0      | 0      | 34         |                  |            |       |          | 0.001136          | 0.00133                    |



22. PCSK9 - chronic kidney disease

ENSG00000169174 | EFO\_0003884

|                              |                          |                                         |
|------------------------------|--------------------------|-----------------------------------------|
| L2G score<br>0.458           | Unmet need index<br>5.24 | SAS global MAF<br>0.001832              |
| Genome India global MAF<br>0 | Variants<br>1            | Interpretations<br>Likely consequential |

Disease Description

The end-stage of CHRONIC RENAL INSUFFICIENCY. It is characterized by the severe irreversible kidney damage (as measured by the level of PROTEINURIA) and the reduction in GLOMERULAR FILTRATION RATE to less than 15 ml per min (Kidney Foundation: Kidney Disease Outcome Quality Initiative, 2002). These patients generally require HEMODIALYSIS or KIDNEY TRANSPLANTATION.

Gene Summary

proprotein convertase subtilisin/kexin type 9

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.458 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Ultra-Orphan

Alleles and Scores

| varid          | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|----------------|------------------|-------------------------|----------|-----------------|--------|--------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 1_55057433_G_C | missense_variant | Likely consequential    | 0.001832 |                 | 0      | 0      | 23.5       | 0.3248            | 0.6684     | 0.674 | 0.2314   | 0.001832          | 0                          |

23. IVNS1ABP - chronic kidney disease

ENSG00000116679 | EFO\_0003884

L2G score  
**0.452**

Unmet need index  
**5.24**

SAS global MAF  
**0.00371**

Genome India global MAF  
**0.003837**

Variants  
**1**

Interpretations  
**Likely consequential**

Disease Description

The end-stage of CHRONIC RENAL INSUFFICIENCY. It is characterized by the severe irreversible kidney damage (as measured by the level of PROTEINURIA) and the reduction in GLOMERULAR FILTRATION RATE to less than 15 ml per min (Kidney Foundation: Kidney Disease Outcome Quality Initiative, 2002). These patients generally require HEMODIALYSIS or KIDNEY TRANSPLANTATION.

Gene Summary

influenza virus NS1A binding protein

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.452 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Ultra-Orphan

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas  | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|---------|-----------------|--------|--------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 1_185298225_T_G | missense_variant | Likely consequential    | 0.00371 | 0.003837        | 0      | 0      | 27.3       | 0.6474            |            | 0.941 | 0.3666   | 0.00371           | 0.003837                   |

24. CYP1A2 - chronic kidney disease

ENSG00000140505 | EFO\_0003884

|                                            |                                 |                                                |
|--------------------------------------------|---------------------------------|------------------------------------------------|
| L2G score<br><b>0.402</b>                  | Unmet need index<br><b>5.24</b> | SAS global MAF<br><b>0.001171</b>              |
| Genome India global MAF<br><b>0.001023</b> | Variants<br><b>1</b>            | Interpretations<br><b>Likely consequential</b> |

Disease Description

The end-stage of CHRONIC RENAL INSUFFICIENCY. It is characterized by the severe irreversible kidney damage (as measured by the level of PROTEINURIA) and the reduction in GLOMERULAR FILTRATION RATE to less than 15 ml per min (Kidney Foundation: Kidney Disease Outcome Quality Initiative, 2002). These patients generally require HEMODIALYSIS or KIDNEY TRANSPLANTATION.

Gene Summary

cytochrome P450 family 1 subfamily A member 2

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.402 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Ultra-Orphan

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_S CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|----------|-----------------|--------|--------|------------|------------------|------------|-------|----------|-------------------|----------------------------|
| 15_74754877_A_T | missense_variant | Likely consequential    | 0.001171 | 0.001023        | 0      | 0      | 23.8       | 0.3768           |            | 0.606 | 0.6782   | 0.001171          | 0.001023                   |

## 25. DAB2 - chronic kidney disease

ENSG00000153071 | EFO\_0003884

|                                            |                                 |                                                |
|--------------------------------------------|---------------------------------|------------------------------------------------|
| L2G score<br><b>0.318</b>                  | Unmet need index<br><b>5.24</b> | SAS global MAF<br><b>0.002203</b>              |
| Genome India global MAF<br><b>0.001637</b> | Variants<br><b>1</b>            | Interpretations<br><b>Likely consequential</b> |

### Disease Description

The end-stage of CHRONIC RENAL INSUFFICIENCY. It is characterized by the severe irreversible kidney damage (as measured by the level of PROTEINURIA) and the reduction in GLOMERULAR FILTRATION RATE to less than 15 ml per min (Kidney Foundation: Kidney Disease Outcome Quality Initiative, 2002). These patients generally require HEMODIALYSIS or KIDNEY TRANSPLANTATION.

### Gene Summary

DAB adaptor protein 2

### Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

### Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.318 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

### SAS / Indian Prevalence

Ultra-Orphan

### Alleles and Scores

| varid          | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe  | AF_fin    | cadd_phred | am_pathogenicity | EVE_S CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|----------------|------------------|-------------------------|----------|-----------------|---------|-----------|------------|------------------|------------|-------|----------|-------------------|----------------------------|
| 5_39394311_C_T | missense_variant | Likely consequential    | 0.002203 | 0.001637        | 0.00187 | 0.0002247 | 28.6       | 0.6487           |            | 0.219 | 0.04278  | 0.002203          | 0.001637                   |

26. DHR511 - chronic kidney disease

ENSG00000278535 | EFO\_0003884

|                              |                          |                                         |
|------------------------------|--------------------------|-----------------------------------------|
| L2G score<br>0.269           | Unmet need index<br>5.24 | SAS global MAF<br>0.001042              |
| Genome India global MAF<br>0 | Variants<br>1            | Interpretations<br>Likely consequential |

Disease Description

The end-stage of CHRONIC RENAL INSUFFICIENCY. It is characterized by the severe irreversible kidney damage (as measured by the level of PROTEINURIA) and the reduction in GLOMERULAR FILTRATION RATE to less than 15 ml per min (Kidney Foundation: Kidney Disease Outcome Quality Initiative, 2002). These patients generally require HEMODIALYSIS or KIDNEY TRANSPLANTATION.

Gene Summary

dehydrogenase/reductase 11

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.269 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Ultra-Orphan

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe   | AF_fin    | cadd_phred | am_pathogenicity | EVE_S CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|----------|-----------------|----------|-----------|------------|------------------|------------|-------|----------|-------------------|----------------------------|
| 17_36595095_T_G | missense_variant | Likely consequential    | 0.001042 |                 | 0.003927 | 0.0003952 | 24.2       | 0.7491           |            |       | 0.9855   | 0.001042          | 0                          |

27. OR52N2 - chronic kidney disease

ENSG00000180988 | EFO\_0003884

L2G score  
**0.218**

Unmet need index  
**5.24**

SAS global MAF  
**0.001204**

Genome India global MAF  
**0.0003582**

Variants  
**1**

Interpretations  
**Likely consequential**

Disease Description

The end-stage of CHRONIC RENAL INSUFFICIENCY. It is characterized by the severe irreversible kidney damage (as measured by the level of PROTEINURIA) and the reduction in GLOMERULAR FILTRATION RATE to less than 15 ml per min (Kidney Foundation: Kidney Disease Outcome Quality Initiative, 2002). These patients generally require HEMODIALYSIS or KIDNEY TRANSPLANTATION.

Gene Summary

olfactory receptor family 52 subfamily N member 2

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.218 across 3 evidence rows and 3 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Ultra-Orphan

Alleles and Scores

| varid          | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_S CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|----------------|------------------|-------------------------|----------|-----------------|--------|--------|------------|------------------|------------|-------|----------|-------------------|----------------------------|
| 11_5820697_C_T | missense_variant | Likely consequential    | 0.001204 | 0.0003582       | 0      | 0      | 21.5       | 0.5659           |            | 0.403 | 0.4332   | 0.001204          | 0.0003582                  |

28. GAB2 - chronic kidney disease

ENSG00000033327 | EFO\_0003884

|                              |                          |                                                          |
|------------------------------|--------------------------|----------------------------------------------------------|
| L2G score<br>0.197           | Unmet need index<br>5.24 | SAS global MAF<br>0.1638                                 |
| Genome India global MAF<br>0 | Variants<br>3            | Interpretations<br>Likely consequential;<br>not_missense |

Disease Description

The end-stage of CHRONIC RENAL INSUFFICIENCY. It is characterized by the severe irreversible kidney damage (as measured by the level of PROTEINURIA) and the reduction in GLOMERULAR FILTRATION RATE to less than 15 ml per min (Kidney Foundation: Kidney Disease Outcome Quality Initiative, 2002). These patients generally require HEMODIALYSIS or KIDNEY TRANSPLANTATION.

Gene Summary

GRB2 associated binding protein 2

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.197 across 2 evidence rows and 2 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Ultra-Orphan

Alleles and Scores

| varid            | consequence        | missense_interpretation | AF_sas   | genome_india_af | AF_nfe    | AF_fin   | cadd_phred | am_pathogenicity | EVE_S CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|------------------|--------------------|-------------------------|----------|-----------------|-----------|----------|------------|------------------|------------|-------|----------|-------------------|----------------------------|
| 11_78226603_T_G  | missense_variant   | Likely consequential    | 0.001572 |                 | 0.0008841 | 0.001251 | 26.3       | 0.9743           |            | 0.251 | 0.3241   | 0.1638            | 0                          |
| 11_78226611_C_G  | missense_variant   | Likely consequential    | 0.1604   |                 | 0.1098    | 0.064    | 24.8       | 0.9349           |            | 0.364 | 0.01756  | 0.1638            | 0                          |
| 11_78226616_T_TG | frameshift_variant | not_misense             | 0.002415 |                 | 0.000905  | 0.001438 | 34         |                  |            |       |          | 0.1638            | 0                          |

29. PSEN1 - intracranial hemorrhage

ENSG00000080815 | EFO\_0000551

L2G score  
**0.751**

Unmet need index  
**5.2**

SAS global MAF  
**0.001961**

Genome India global MAF  
**0**

Variants  
**1**

Interpretations  
**Likely consequential**

Disease Description

Bleeding within the SKULL, including hemorrhages in the brain and the three membranes of MENINGES. The escape of blood often leads to the formation of HEMATOMA in the cranial epidural, subdural, and subarachnoid spaces.

Gene Summary

| presenilin 1

Known Drugs / Clinical Investigation

| No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.421, evidence 1; europepmc: score 0.0517, evidence 2. GWAS credible-set evidence in this analysis has L2G 0.751, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Common

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe   | AF_fin    | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|----------|-----------------|----------|-----------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 14_73192774_A_C | missense_variant | Likely consequential    | 0.001961 |                 | 0.007899 | 0.0009889 | 25.1       | 0.611             | 0.6764     | 0.941 | 0.9803   | 0.001961          | 0                          |



30. EXOC4 - intracranial hemorrhage

ENSG00000131558 | EFO\_0000551

L2G score  
**0.711**

Unmet need index  
**5.2**

SAS global MAF  
**0.007772**

Genome India global MAF  
**0.008239**

Variants  
**1**

Interpretations  
**Likely consequential**

Disease Description

Bleeding within the SKULL, including hemorrhages in the brain and the three membranes of MENINGES. The escape of blood often leads to the formation of HEMATOMA in the cranial epidural, subdural, and subarachnoid spaces.

Gene Summary

exocyst complex component 4

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.459, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.711, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Common

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe   | AF_fin   | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|----------|-----------------|----------|----------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 7_134064365_C_T | missense_variant | Likely consequential    | 0.007772 | 0.008239        | 0.001528 | 0.000612 | 29.2       | 0.4057            | 0.7702     | 0.367 | 0.03259  | 0.007772          | 0.008239                   |

31. ANKS1B - intracranial hemorrhage

ENSG00000185046 | EFO\_0000551

|                                     |                         |                                         |
|-------------------------------------|-------------------------|-----------------------------------------|
| L2G score<br>0.621                  | Unmet need index<br>5.2 | SAS global MAF<br>8.115e-05             |
| Genome India global MAF<br>0.001228 | Variants<br>1           | Interpretations<br>Likely consequential |

Disease Description

Bleeding within the SKULL, including hemorrhages in the brain and the three membranes of MENINGES. The escape of blood often leads to the formation of HEMATOMA in the cranial epidural, subdural, and subarachnoid spaces.

Gene Summary

ankyrin repeat and sterile alpha motif domain containing 1B

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.421, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.621, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Common

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas    | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_S CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|-----------|-----------------|--------|--------|------------|------------------|------------|-------|----------|-------------------|----------------------------|
| 12_99806627_T_C | missense_variant | Likely consequential    | 8.115e-05 | 0.001228        | 0      | 0      | 27.7       | 0.2076           |            |       | 0.7108   | 8.115e-05         | 0.001228                   |

32. WWOX - intracranial hemorrhage

ENSG00000186153 | EFO\_0000551

|                                           |                                |                                                |
|-------------------------------------------|--------------------------------|------------------------------------------------|
| L2G score<br><b>0.611</b>                 | Unmet need index<br><b>5.2</b> | SAS global MAF<br><b>0.08356</b>               |
| Genome India global MAF<br><b>0.09746</b> | Variants<br><b>2</b>           | Interpretations<br><b>Likely consequential</b> |

Disease Description

Bleeding within the SKULL, including hemorrhages in the brain and the three membranes of MENINGES. The escape of blood often leads to the formation of HEMATOMA in the cranial epidural, subdural, and subarachnoid spaces.

Gene Summary

WW domain containing oxidoreductase

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.394, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.611, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Common

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas  | genome_india_af | AF_nfe    | AF_fin   | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|---------|-----------------|-----------|----------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 16_78115103_C_T | missense_variant | Likely consequential    | 0.01693 | 0.01202         | 0.007544  | 0.01005  | 31         | 0.1905            | 0.5718     |       | 0.02518  | 0.08356           | 0.09746                    |
| 16_78425018_C_G | missense_variant | Likely consequential    | 0.06778 | 0.08648         | 0.0002878 | 0.002808 | 24         | 0.1893            | 0.5105     |       | 0.02795  | 0.08356           | 0.09746                    |

33. ACACA - intracranial hemorrhage

ENSG00000278540 | EFO\_0000551

L2G score  
**0.594**

Unmet need index  
**5.2**

SAS global MAF  
**0.001516**

Genome India global MAF  
**0**

Variants  
**1**

Interpretations  
**Likely consequential**

Disease Description

Bleeding within the SKULL, including hemorrhages in the brain and the three membranes of MENINGES. The escape of blood often leads to the formation of HEMATOMA in the cranial epidural, subdural, and subarachnoid spaces.

Gene Summary

acetyl-CoA carboxylase alpha

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.346, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.594, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Common

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe   | AF_fin    | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|----------|-----------------|----------|-----------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 17_37221733_T_G | missense_variant | Likely consequential    | 0.001516 |                 | 0.005953 | 0.0001697 | 25.7       |                   |            |       | 0.9725   | 0.001516          | 0                          |

34. PSPH - intracranial hemorrhage

ENSG00000146733 | EFO\_0000551

|                                            |                                |                                                                  |
|--------------------------------------------|--------------------------------|------------------------------------------------------------------|
| L2G score<br><b>0.504</b>                  | Unmet need index<br><b>5.2</b> | SAS global MAF<br><b>0.5026</b>                                  |
| Genome India global MAF<br><b>0.000307</b> | Variants<br><b>6</b>           | Interpretations<br><b>Likely consequential;<br/>not_missense</b> |

Disease Description

Bleeding within the SKULL, including hemorrhages in the brain and the three membranes of MENINGES. The escape of blood often leads to the formation of HEMATOMA in the cranial epidural, subdural, and subarachnoid spaces.

Gene Summary

| phosphoserine phosphatase

Known Drugs / Clinical Investigation

| No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.291, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.504, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Common

Alleles and Scores

| varid           | consequence          | missense_interpretation | AF_sas  | genome_india_af | AF_nfe  | AF_fin  | cadd_phred | am_pathogenicity | EVE_S CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|----------------------|-------------------------|---------|-----------------|---------|---------|------------|------------------|------------|-------|----------|-------------------|----------------------------|
| 7_56019598_AC_A | splice_donor_variant | not_missense            | 0.05696 |                 | 0.0123  | 0.07816 | 27.6       |                  |            |       |          | 0.5026            | 0.000307                   |
| 7_56019607_C_T  | missense_variant     | Likely consequential    | 0.06794 | 0.000307        | 0.01486 | 0.08684 | 24.7       | 0.4362           | 0.3032     | 0.738 | 0.1532   | 0.5026            | 0.000307                   |
| 7_56019672_A_G  | missense_variant     | Likely consequential    | 0.1325  |                 | 0.01804 | 0.09756 | 26.5       | 0.9648           | 0.415      | 0.748 | 0.1037   | 0.5026            | 0.000307                   |
| 7_56019681_C_T  | missense_variant     | Likely consequential    | 0.1195  |                 | 0.01489 | 0.09731 | 28.3       | 0.9682           | 0.7713     | 0.818 | 0.1607   | 0.5026            | 0.000307                   |
| 7_56019730_G_A  | missense_variant     | Likely consequential    | 0.1481  |                 | 0.02094 | 0.1173  | 22.1       | 0.3969           | 0.4362     | 0.595 | 0.03455  | 0.5026            | 0.000307                   |

| varid        | consequence      | missense_interpretation | AF_sas | genome_india_af | AF_nfe  | AF_fin | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|--------------|------------------|-------------------------|--------|-----------------|---------|--------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 7_5602118_TC | missense_variant | Likely consequential    | 0.1304 |                 | 0.03035 | 0.1151 | 33         | 0.9506           | 0.7497    | 0.692 | 0.06746  | 0.5026            | 0.000307                   |

35. TMEM130 - intracranial hemorrhage

ENSG00000166448 | EFO\_0000551

L2G score  
**0.395**

Unmet need index  
**5.2**

SAS global MAF  
**0.001731**

Genome India global MAF  
**0.001995**

Variants  
**1**

Interpretations  
**Likely consequential**

Disease Description

Bleeding within the SKULL, including hemorrhages in the brain and the three membranes of MENINGES. The escape of blood often leads to the formation of HEMATOMA in the cranial epidural, subdural, and subarachnoid spaces.

Gene Summary

transmembrane protein 130

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.173, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.395, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Common

Alleles and Scores

| varid          | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_S CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|----------------|------------------|-------------------------|----------|-----------------|--------|--------|------------|------------------|------------|-------|----------|-------------------|----------------------------|
| 7_98851615_A_G | missense_variant | Likely consequential    | 0.001731 | 0.001995        | 0      | 0      | 26.4       | 0.8845           |            | 0.305 | 0.09118  | 0.001731          | 0.001995                   |

### 36. RTEL1 - glioblastoma multiforme

ENSG00000258366 | EFO\_0000519

|                                             |                                 |                                                |
|---------------------------------------------|---------------------------------|------------------------------------------------|
| L2G score<br><b>0.451</b>                   | Unmet need index<br><b>5.07</b> | SAS global MAF<br><b>0.001287</b>              |
| Genome India global MAF<br><b>0.0008698</b> | Variants<br><b>1</b>            | Interpretations<br><b>Likely consequential</b> |

#### Disease Description

The most malignant astrocytic tumor (WHO grade IV). It is composed of poorly differentiated neoplastic astrocytes and it is characterized by the presence of cellular polymorphism, nuclear atypia, brisk mitotic activity, vascular thrombosis, microvascular proliferation and necrosis. It typically affects adults and is preferentially located in the cerebral hemispheres. It may develop from diffuse astrocytoma WHO grade II or anaplastic astrocytoma (secondary glioblastoma, IDH-mutant), but more frequently, it manifests after a short clinical history de novo, without evidence of a less malignant precursor lesion (primary glioblastoma, IDH- wildtype). (Adapted from WHO)

#### Gene Summary

regulator of telomere elongation helicase 1

#### Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

#### Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.451 across 3 evidence rows and 3 study loci. No datasource-level direct association summary was found in the downloaded association table.

#### SAS / Indian Prevalence

Ultra-Orphan

#### Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|----------|-----------------|--------|--------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 20_63662850_G_A | missense_variant | Likely consequential    | 0.001287 | 0.0008698       | 0      | 0      | 25.2       | 0.6091           | 0.3284    | 0.519 | 0.3006   | 0.001287          | 0.0008698                  |



### 37. HNF1B - ovarian carcinoma

ENSG00000275410 | EFO\_0001075

|                                     |                                 |                                        |
|-------------------------------------|---------------------------------|----------------------------------------|
| L2G score<br><b>0.865</b>           | Unmet need index<br><b>5.07</b> | SAS global MAF<br><b>0.0169</b>        |
| Genome India global MAF<br><b>0</b> | Variants<br><b>1</b>            | Interpretations<br><b>not_missense</b> |

#### Disease Description

A malignant neoplasm originating from the surface ovarian epithelium. It accounts for the greatest number of deaths from malignancies of the female genital tract and is the fifth leading cause of cancer fatalities in women. It is predominantly a disease of older white women of northern European extraction, but it is seen in all ages and ethnic groups. Adenocarcinomas constitute the vast majority of ovarian carcinomas. The pattern of metastatic spread in ovarian carcinoma is similar regardless of the microscopic type. The most common sites of involvement are the contralateral ovary, peritoneal cavity, para-aortic and pelvic lymph nodes, and liver. Lung and pleura are the most common sites of extra-abdominal spread. The primary form of therapy is surgical. The overall prognosis of ovarian carcinoma remains poor, a direct result of its rapid growth rate and the lack of early symptoms. --2002

#### Gene Summary

HNF1 homeobox B

#### Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

#### Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.865 across 4 evidence rows and 4 study loci. No datasource-level direct association summary was found in the downloaded association table.

#### SAS / Indian Prevalence

Rare

#### Alleles and Scores

| varid                  | consequence        | missense_interpretation | AF_sas | genome_india_af | AF_nfe | AF_fin  | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|------------------------|--------------------|-------------------------|--------|-----------------|--------|---------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 17_37710582_GTCACCAT_G | frameshift_variant | not_missense            | 0.0169 |                 | 0.0023 | 0.02255 | 35         |                  |           |       |          | 0.0169            | 0                          |

38. ABO - ovarian carcinoma

ENSG00000175164 | EFO\_0001075

|                                    |                          |                                 |
|------------------------------------|--------------------------|---------------------------------|
| L2G score<br>0.799                 | Unmet need index<br>5.07 | SAS global MAF<br>0.04705       |
| Genome India global MAF<br>0.03973 | Variants<br>2            | Interpretations<br>not_missense |

Disease Description

A malignant neoplasm originating from the surface ovarian epithelium. It accounts for the greatest number of deaths from malignancies of the female genital tract and is the fifth leading cause of cancer fatalities in women. It is predominantly a disease of older white women of northern European extraction, but it is seen in all ages and ethnic groups. Adenocarcinomas constitute the vast majority of ovarian carcinomas. The pattern of metastatic spread in ovarian carcinoma is similar regardless of the microscopic type. The most common sites of involvement are the contralateral ovary, peritoneal cavity, para-aortic and pelvic lymph nodes, and liver. Lung and pleura are the most common sites of extra-abdominal spread. The primary form of therapy is surgical. The overall prognosis of ovarian carcinoma remains poor, a direct result of its rapid growth rate and the lack of early symptoms. --2002

Gene Summary

ABO, alpha 1-3-N-acetylgalactosaminyltransferase and alpha 1-3-galactosyltransferase

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.799 across 4 evidence rows and 4 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid             | consequence        | missense_interpretation | AF_sas  | genome_india_af | AF_nfe   | AF_fin    | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-------------------|--------------------|-------------------------|---------|-----------------|----------|-----------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 9_133256189_C_T   | stop_gained        | not_missense            | 0.0329  | 0.02446         | 0.01796  | 0.01162   | 51         |                  |           |       |          | 0.04705           | 0.03973                    |
| 9_133256234_G_T_G | frameshift_variant | not_missense            | 0.01463 | 0.01566         | 0.002961 | 0.0003027 | 17.65      |                  |           |       |          | 0.04705           | 0.03973                    |

39. HOXD3 - ovarian carcinoma

ENSG00000128652 | EFO\_0001075

|                                            |                                 |                                                |
|--------------------------------------------|---------------------------------|------------------------------------------------|
| L2G score<br><b>0.739</b>                  | Unmet need index<br><b>5.07</b> | SAS global MAF<br><b>0.0005333</b>             |
| Genome India global MAF<br><b>0.001023</b> | Variants<br><b>1</b>            | Interpretations<br><b>Likely consequential</b> |

Disease Description

A malignant neoplasm originating from the surface ovarian epithelium. It accounts for the greatest number of deaths from malignancies of the female genital tract and is the fifth leading cause of cancer fatalities in women. It is predominantly a disease of older white women of northern European extraction, but it is seen in all ages and ethnic groups. Adenocarcinomas constitute the vast majority of ovarian carcinomas. The pattern of metastatic spread in ovarian carcinoma is similar regardless of the microscopic type. The most common sites of involvement are the contralateral ovary, peritoneal cavity, para-aortic and pelvic lymph nodes, and liver. Lung and pleura are the most common sites of extra-abdominal spread. The primary form of therapy is surgical. The overall prognosis of ovarian carcinoma remains poor, a direct result of its rapid growth rate and the lack of early symptoms. --2002

Gene Summary

homeobox D3

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.739 across 5 evidence rows and 5 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|----------|-----------------|--------|--------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 2_176172148_C_G | missense_variant | Likely consequential    | 0.000533 | 0.001023        | 0      | 0      | 26.2       | 0.8374           |           |       | 0.4206   | 0.000533          | 0.001023                   |

40. POU5F1B - ovarian carcinoma

ENSG00000212993 | EFO\_0001075

|                                            |                                 |                                                                  |
|--------------------------------------------|---------------------------------|------------------------------------------------------------------|
| L2G score<br><b>0.54</b>                   | Unmet need index<br><b>5.07</b> | SAS global MAF<br><b>0.001041</b>                                |
| Genome India global MAF<br><b>0.002812</b> | Variants<br><b>2</b>            | Interpretations<br><b>Likely consequential;<br/>not_missense</b> |

Disease Description

A malignant neoplasm originating from the surface ovarian epithelium. It accounts for the greatest number of deaths from malignancies of the female genital tract and is the fifth leading cause of cancer fatalities in women. It is predominantly a disease of older white women of northern European extraction, but it is seen in all ages and ethnic groups. Adenocarcinomas constitute the vast majority of ovarian carcinomas. The pattern of metastatic spread in ovarian carcinoma is similar regardless of the microscopic type. The most common sites of involvement are the contralateral ovary, peritoneal cavity, para-aortic and pelvic lymph nodes, and liver. Lung and pleura are the most common sites of extra-abdominal spread. The primary form of therapy is surgical. The overall prognosis of ovarian carcinoma remains poor, a direct result of its rapid growth rate and the lack of early symptoms. --2002

Gene Summary

POU class 5 homeobox 1B

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.54 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas    | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|-----------|-----------------|--------|--------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 8_127416359_G_C | missense_variant | Likely consequential    | 0.0008054 | 0.001586        | 0      | 0      | 20.9       | 0.6828           |           |       | 0.9479   | 0.001041          | 0.002812                   |
| 8_127416785_G_T | stop_gained      | not_misense             | 0.0002362 | 0.001228        | 0      | 0      | 35         |                  |           |       |          | 0.001041          | 0.002812                   |

41. ATP6V0A1 - ovarian carcinoma

ENSG00000033627 | EFO\_0001075

|                                     |                          |                                         |
|-------------------------------------|--------------------------|-----------------------------------------|
| L2G score<br>0.357                  | Unmet need index<br>5.07 | SAS global MAF<br>0.00021               |
| Genome India global MAF<br>0.001433 | Variants<br>1            | Interpretations<br>Likely consequential |

Disease Description

A malignant neoplasm originating from the surface ovarian epithelium. It accounts for the greatest number of deaths from malignancies of the female genital tract and is the fifth leading cause of cancer fatalities in women. It is predominantly a disease of older white women of northern European extraction, but it is seen in all ages and ethnic groups. Adenocarcinomas constitute the vast majority of ovarian carcinomas. The pattern of metastatic spread in ovarian carcinoma is similar regardless of the microscopic type. The most common sites of involvement are the contralateral ovary, peritoneal cavity, para-aortic and pelvic lymph nodes, and liver. Lung and pleura are the most common sites of extra-abdominal spread. The primary form of therapy is surgical. The overall prognosis of ovarian carcinoma remains poor, a direct result of its rapid growth rate and the lack of early symptoms. --2002

Gene Summary

ATPase H+ transporting V0 subunit a1

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.357 across 4 evidence rows and 4 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas  | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|---------|-----------------|--------|--------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 17_42521077_C_G | missense_variant | Likely consequential    | 0.00021 | 0.001433        | 0      | 0      | 29.3       | 0.9754            |            | 0.97  | 0.9982   | 0.00021           | 0.001433                   |

42. SPPL2C - ovarian carcinoma

ENSG00000185294 | EFO\_0001075

|                                           |                                 |                                                |
|-------------------------------------------|---------------------------------|------------------------------------------------|
| L2G score<br><b>0.234</b>                 | Unmet need index<br><b>5.07</b> | SAS global MAF<br><b>0.07968</b>               |
| Genome India global MAF<br><b>0.05546</b> | Variants<br><b>1</b>            | Interpretations<br><b>Likely consequential</b> |

Disease Description

A malignant neoplasm originating from the surface ovarian epithelium. It accounts for the greatest number of deaths from malignancies of the female genital tract and is the fifth leading cause of cancer fatalities in women. It is predominantly a disease of older white women of northern European extraction, but it is seen in all ages and ethnic groups. Adenocarcinomas constitute the vast majority of ovarian carcinomas. The pattern of metastatic spread in ovarian carcinoma is similar regardless of the microscopic type. The most common sites of involvement are the contralateral ovary, peritoneal cavity, para-aortic and pelvic lymph nodes, and liver. Lung and pleura are the most common sites of extra-abdominal spread. The primary form of therapy is surgical. The overall prognosis of ovarian carcinoma remains poor, a direct result of its rapid growth rate and the lack of early symptoms. --2002

Gene Summary

| signal peptide peptidase like 2C

Known Drugs / Clinical Investigation

| No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.234 across 6 evidence rows and 6 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas  | genome_india_af | AF_nfe | AF_fin  | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|---------|-----------------|--------|---------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 17_45846288_G_C | missense_variant | Likely consequential    | 0.07968 | 0.05546         | 0.2218 | 0.07211 | 23.2       | 0.8506           |           | 0.293 | 0.1048   | 0.07968           | 0.05546                    |

43. MAST4 - bullous pemphigoid

ENSG00000069020 | EFO\_0007187

|                                      |                          |                                 |
|--------------------------------------|--------------------------|---------------------------------|
| L2G score<br>0.871                   | Unmet need index<br>5.07 | SAS global MAF<br>0.001132      |
| Genome India global MAF<br>0.0008698 | Variants<br>1            | Interpretations<br>not_missense |

Disease Description

Bullous pemphigoid (BP) is the most common form of autoimmune bullous dermatosis.

Gene Summary

microtubule associated serine/threonine kinase family member 4

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.871 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid           | consequence        | missense_interpretation | AF_sas   | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|--------------------|-------------------------|----------|-----------------|--------|--------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 5_67166133_GA_G | frameshift_variant | not_missense            | 0.001132 | 0.0008698       | 0      | 0      | 24         |                  |           |       |          | 0.001132          | 0.0008698                  |

44. PARD3 - cardiac arrest

ENSG00000148498 | EFO\_0009492

L2G score  
**0.877**

Unmet need index  
**5.03**

SAS global MAF  
**0.0003711**

Genome India global MAF  
**0.001586**

Variants  
**1**

Interpretations  
**Likely consequential**

Disease Description

Cessation of breathing and/or cardiac function.

Gene Summary

| par-3 family cell polarity regulator

Known Drugs / Clinical Investigation

| No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.625, evidence 2. GWAS credible-set evidence in this analysis has L2G 0.877, 2 evidence rows, and 2 study loci.

SAS / Indian Prevalence

Ultra-Orphan

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas    | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_S CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|-----------|-----------------|--------|--------|------------|------------------|------------|-------|----------|-------------------|----------------------------|
| 10_34341687_C_T | missense_variant | Likely consequential    | 0.0003711 | 0.001586        | 0      | 0      | 24.4       |                  |            | 0.124 | 0.7087   | 0.0003711         | 0.001586                   |



45. IGSF5 - cardiac arrest

ENSG00000183067 | EFO\_0009492

|                                     |                          |                                         |
|-------------------------------------|--------------------------|-----------------------------------------|
| L2G score<br>0.758                  | Unmet need index<br>5.03 | SAS global MAF<br>0.0007783             |
| Genome India global MAF<br>0.001279 | Variants<br>1            | Interpretations<br>Likely consequential |

Disease Description

Cessation of breathing and/or cardiac function.

Gene Summary

immunoglobulin superfamily member 5

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.461, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.758, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Ultra-Orphan

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas    | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|-----------|-----------------|--------|--------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 21_39765840_C_T | missense_variant | Likely consequential    | 0.0007783 | 0.001279        | 0      | 0      | 23.9       | 0.7657           |           | 0.362 | 0.5046   | 0.0007783         | 0.001279                   |

46. THSD7B - cardiac arrest

ENSG00000144229 | EFO\_0009492

|                                    |                          |                                 |
|------------------------------------|--------------------------|---------------------------------|
| L2G score<br>0.752                 | Unmet need index<br>5.03 | SAS global MAF<br>0.04541       |
| Genome India global MAF<br>0.05993 | Variants<br>1            | Interpretations<br>not_missense |

Disease Description

Cessation of breathing and/or cardiac function.

Gene Summary

thrombospondin type 1 domain containing 7B

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.502, evidence 2. GWAS credible-set evidence in this analysis has L2G 0.752, 2 evidence rows, and 2 study loci.

SAS / Indian Prevalence

Ultra-Orphan

Alleles and Scores

| varid           | consequence          | missense_interpretation | AF_sas  | genome_india_af | AF_nfe   | AF_fin   | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|----------------------|-------------------------|---------|-----------------|----------|----------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 2_137272664_C_T | splice_donor_variant | not_missense            | 0.04541 | 0.05993         | 0.007994 | 0.009018 | 20.5       |                  |           |       |          | 0.04541           | 0.05993                    |

47. PREX2 - cardiac arrest

ENSG000000046889 | EFO\_0009492

L2G score  
**0.726**

Unmet need index  
**5.03**

SAS global MAF  
**0.006294**

Genome India global MAF  
**0.0003582**

Variants  
**2**

Interpretations  
**Likely consequential**

Disease Description

Cessation of breathing and/or cardiac function.

Gene Summary

| phosphatidylinositol-3,4,5-trisphosphate dependent Rac exchange factor 2

Known Drugs / Clinical Investigation

| No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.485, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.726, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Ultra-Orphan

Alleles and Scores

| varid          | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe    | AF_fin    | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|----------------|------------------|-------------------------|----------|-----------------|-----------|-----------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 8_68044581_G_A | missense_variant | Likely consequential    | 0.001104 | 0.0003582       | 0.002941  | 0.0001686 | 27.1       | 0.6788           | 0.5794    | 0.636 | 0.05465  | 0.006294          | 0.0003582                  |
| 8_68192488_G_C | missense_variant | Likely consequential    | 0.005195 |                 | 0.0009588 | 0.01402   | 23.3       | 0.9207           | 0.4299    | 0.635 | 0.04875  | 0.006294          | 0.0003582                  |

48. FMN2 - cardiac arrest

ENSG00000155816 | EFO\_0009492

|                              |                          |                                 |
|------------------------------|--------------------------|---------------------------------|
| L2G score<br>0.69            | Unmet need index<br>5.03 | SAS global MAF<br>0.03854       |
| Genome India global MAF<br>0 | Variants<br>3            | Interpretations<br>not_missense |

Disease Description

Cessation of breathing and/or cardiac function.

Gene Summary

formin 2

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.479, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.69, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Ultra-Orphan

Alleles and Scores

| varid            | consequence        | missense_interpretation | AF_sas  | genome_india_af | AF_nfe   | AF_fin  | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|------------------|--------------------|-------------------------|---------|-----------------|----------|---------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 1_240092269_GC_G | frameshift_variant | not_missense            | 0.01521 |                 | 0.003124 | 0.03017 | 27.3       |                  |           |       |          | 0.03854           | 0                          |
| 1_240207664_CA   | frameshift_variant | not_missense            | 0.01078 |                 | 0.009574 | 0.04247 | 21.5       |                  |           |       |          | 0.03854           | 0                          |
| 1_240207667_CA_C | frameshift_variant | not_missense            | 0.01306 |                 | 0.01156  | 0.05875 | 14.43      |                  |           |       |          | 0.03854           | 0                          |

49. ADCY5 - cardiac arrest

ENSG00000173175 | EFO\_0009492

|                                      |                          |                                         |
|--------------------------------------|--------------------------|-----------------------------------------|
| L2G score<br>0.641                   | Unmet need index<br>5.03 | SAS global MAF<br>0.00313               |
| Genome India global MAF<br>0.0007163 | Variants<br>2            | Interpretations<br>Likely consequential |

Disease Description

Cessation of breathing and/or cardiac function.

Gene Summary

adenylate cyclase 5

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.337, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.641, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Ultra-Orphan

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe   | AF_fin   | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|----------|-----------------|----------|----------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 3_123300202_C_T | missense_variant | Likely consequential    | 0.001982 | 0.0007163       | 0.000893 | 0.002507 | 25.4       | 0.736            | 0.6324    | 0.45  | 0.02516  | 0.00313           | 0.0007163                  |
| 3_123447564_C_A | missense_variant | Likely consequential    | 0.00115  |                 | 0        | 0        | 29.6       | 0.9746           | 0.9394    | 0.377 | 0.7995   | 0.00313           | 0.0007163                  |

50. SPDYE21 - cardiac arrest

ENSG00000230358 | EFO\_0009492

|                                   |                          |                                 |
|-----------------------------------|--------------------------|---------------------------------|
| L2G score<br>0.539                | Unmet need index<br>5.03 | SAS global MAF<br>0.1466        |
| Genome India global MAF<br>0.2538 | Variants<br>2            | Interpretations<br>not_missense |

Disease Description

Cessation of breathing and/or cardiac function.

Gene Summary

| speedy/RINGO cell cycle regulator family member E21

Known Drugs / Clinical Investigation

| No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.364, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.539, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Ultra-Orphan

Alleles and Scores

| varid                 | consequence          | missense_interpretation | AF_sas | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------------|----------------------|-------------------------|--------|-----------------|--------|--------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 7_67278874_G_A        | splice_donor_variant | not_misense             |        | 0.08781         |        |        | 17.61      |                  |           |       |          | 0.1466            | 0.2538                     |
| 7_67281505_T_TGGGTGGC | frameshift_variant   | not_misense             | 0.1466 | 0.182           | 0.1583 | 0.2605 | 22.2       |                  |           |       |          | 0.1466            | 0.2538                     |

51. OR10AD1 - cardiac arrest

ENSG00000172640 | EFO\_0009492

|                                     |                          |                                 |
|-------------------------------------|--------------------------|---------------------------------|
| L2G score<br>0.226                  | Unmet need index<br>5.03 | SAS global MAF<br>0.002723      |
| Genome India global MAF<br>0.002507 | Variants<br>1            | Interpretations<br>not_missense |

Disease Description

Cessation of breathing and/or cardiac function.

Gene Summary

olfactory receptor family 10 subfamily AD member 1

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.152, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.226, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Ultra-Orphan

Alleles and Scores

| varid           | consequence | missense_interpretation | AF_sas   | genome_india_af | AF_nfe   | AF_fin    | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|-------------|-------------------------|----------|-----------------|----------|-----------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 12_48202807_A_C | stop_gained | not_missense            | 0.002723 | 0.002507        | 0.001946 | 0.0007715 | 33         |                  |           |       |          | 0.002723          | 0.002507                   |

52. LMTK3 - cardiac arrest

ENSG00000142235 | EFO\_0009492

L2G score  
**0.214**

Unmet need index  
**5.03**

SAS global MAF  
**0.02655**

Genome India global MAF  
**0**

Variants  
**4**

Interpretations  
**Likely consequential;  
not\_missense**

Disease Description

Cessation of breathing and/or cardiac function.

Gene Summary

lemur tyrosine kinase 3

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.186, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.214, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Ultra-Orphan

Alleles and Scores

| varid              | consequence        | missense_interpretation | AF_sas   | genome_india_af | AF_nfe    | AF_fin   | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|--------------------|--------------------|-------------------------|----------|-----------------|-----------|----------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 19_48491500_C_CG   | frameshift_variant | not_missense            | 0.001048 |                 | 0.0001579 | 0.001178 | 35         |                  |           |       |          | 0.02655           | 0                          |
| 19_48498801_C_GG_C | frameshift_variant | not_missense            | 0.01424  |                 | 0.004306  | 0.001102 | 33         |                  |           |       |          | 0.02655           | 0                          |
| 19_48498801_C_CG   | frameshift_variant | not_missense            | 0.006912 |                 | 0.01321   | 0.005314 | 29.4       |                  |           |       |          | 0.02655           | 0                          |
| 19_48509445_A_C    | missense_variant   | Likely consequential    | 0.004569 |                 | 0.001078  | 0.008125 | 26.2       | 0.996            |           |       | 0.03098  | 0.02655           | 0                          |



53. TP63 - lung adenocarcinoma

ENSG00000073282 | EFO\_0000571

L2G score  
0.906

Unmet need index  
5

SAS global MAF  
0.001565

Genome India global MAF  
0.001433

Variants  
1

Interpretations  
Likely consequential

Disease Description

A carcinoma that arises from the lung and is characterized by the presence of malignant glandular epithelial cells. There is a male predilection with a male to female ratio of 2:1. Usually lung adenocarcinoma is asymptomatic and is identified through screening studies or as an incidental radiologic finding. If clinical symptoms are present they include shortness of breath, cough, hemoptysis, chest pain, and fever. Tobacco smoke is a known risk factor.

Gene Summary

tumor protein p63

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.906 across 1e+01 evidence rows and 1e+01 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Uncommon

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe   | AF_fin    | cadd_phred | am_pathogenicity | EVE_S CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|----------|-----------------|----------|-----------|------------|------------------|------------|-------|----------|-------------------|----------------------------|
| 3_189889363_C_A | missense_variant | Likely consequential    | 0.001565 | 0.001433        | 0.003678 | 0.0008985 | 24.6       | 0.0744           | 0.2006     | 0.553 | 0.02005  | 0.001565          | 0.001433                   |

54. SECISBP2L - lung adenocarcinoma

ENSG00000138593 | EFO\_0000571

|                                     |                       |                                         |
|-------------------------------------|-----------------------|-----------------------------------------|
| L2G score<br>0.884                  | Unmet need index<br>5 | SAS global MAF<br>0.0002435             |
| Genome India global MAF<br>0.001279 | Variants<br>1         | Interpretations<br>Likely consequential |

Disease Description

A carcinoma that arises from the lung and is characterized by the presence of malignant glandular epithelial cells. There is a male predilection with a male to female ratio of 2:1. Usually lung adenocarcinoma is asymptomatic and is identified through screening studies or as an incidental radiologic finding. If clinical symptoms are present they include shortness of breath, cough, hemoptysis, chest pain, and fever. Tobacco smoke is a known risk factor.

Gene Summary

SECIS binding protein 2 like

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.884 across 6 evidence rows and 6 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Uncommon

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas    | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|-----------|-----------------|--------|--------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 15_48992719_T_C | missense_variant | Likely consequential    | 0.0002435 | 0.001279        | 0      | 0      | 22.7       | 0.1125            |            |       | 0.597    | 0.0002435         | 0.001279                   |

55. HNF1B - lung adenocarcinoma

ENSG00000275410 | EFO\_0000571

|                              |                       |                                 |
|------------------------------|-----------------------|---------------------------------|
| L2G score<br>0.865           | Unmet need index<br>5 | SAS global MAF<br>0.0169        |
| Genome India global MAF<br>0 | Variants<br>1         | Interpretations<br>not_missense |

Disease Description

A carcinoma that arises from the lung and is characterized by the presence of malignant glandular epithelial cells. There is a male predilection with a male to female ratio of 2:1. Usually lung adenocarcinoma is asymptomatic and is identified through screening studies or as an incidental radiologic finding. If clinical symptoms are present they include shortness of breath, cough, hemoptysis, chest pain, and fever. Tobacco smoke is a known risk factor.

Gene Summary

HNF1 homeobox B

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.865 across 2 evidence rows and 2 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Uncommon

Alleles and Scores

| varid                  | consequence        | missense_interpretation | AF_sas | genome_india_af | AF_nfe | AF_fin  | cadd_phred | am_pathogenicity | EVE_S CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|------------------------|--------------------|-------------------------|--------|-----------------|--------|---------|------------|------------------|------------|-------|----------|-------------------|----------------------------|
| 17_37710582_GTCACCAT_G | frameshift_variant | not_missense            | 0.0169 |                 | 0.0023 | 0.02255 | 35         |                  |            |       |          | 0.0169            | 0                          |

56. FOXP4 - lung adenocarcinoma

ENSG00000137166 | EFO\_0000571

|                                            |                              |                                                |
|--------------------------------------------|------------------------------|------------------------------------------------|
| L2G score<br><b>0.575</b>                  | Unmet need index<br><b>5</b> | SAS global MAF<br><b>0.001021</b>              |
| Genome India global MAF<br><b>0.001279</b> | Variants<br><b>1</b>         | Interpretations<br><b>Likely consequential</b> |

Disease Description

A carcinoma that arises from the lung and is characterized by the presence of malignant glandular epithelial cells. There is a male predilection with a male to female ratio of 2:1. Usually lung adenocarcinoma is asymptomatic and is identified through screening studies or as an incidental radiologic finding. If clinical symptoms are present they include shortness of breath, cough, hemoptysis, chest pain, and fever. Tobacco smoke is a known risk factor.

Gene Summary

forkhead box P4

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.575 across 4 evidence rows and 4 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Uncommon

Alleles and Scores

| varid         | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|---------------|------------------|-------------------------|----------|-----------------|--------|--------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 6_41597932_TC | missense_variant | Likely consequential    | 0.001021 | 0.001279        | 0      | 0      | 25.8       | 0.1071            |            | 0.608 | 0.1747   | 0.001021          | 0.001279                   |

57. TCF7L2 - lung adenocarcinoma

ENSG00000148737 | EFO\_0000571

|                              |                       |                                         |
|------------------------------|-----------------------|-----------------------------------------|
| L2G score<br>0.435           | Unmet need index<br>5 | SAS global MAF<br>0.002843              |
| Genome India global MAF<br>0 | Variants<br>1         | Interpretations<br>Likely consequential |

Disease Description

A carcinoma that arises from the lung and is characterized by the presence of malignant glandular epithelial cells. There is a male predilection with a male to female ratio of 2:1. Usually lung adenocarcinoma is asymptomatic and is identified through screening studies or as an incidental radiologic finding. If clinical symptoms are present they include shortness of breath, cough, hemoptysis, chest pain, and fever. Tobacco smoke is a known risk factor.

Gene Summary

transcription factor 7 like 2

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.435 across 3 evidence rows and 3 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Uncommon

Alleles and Scores

| varid            | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe   | AF_fin   | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|------------------|------------------|-------------------------|----------|-----------------|----------|----------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 10_113165649_T_C | missense_variant | Likely consequential    | 0.002843 |                 | 0.001924 | 0.002653 | 26.4       | 0.0881            | 0.0369     | 0.521 | 0.03355  | 0.002843          | 0                          |

58. RTEL1 - lung adenocarcinoma

ENSG00000258366 | EFO\_0000571

L2G score  
**0.277**

Unmet need index  
**5**

SAS global MAF  
**0.001287**

Genome India global MAF  
**0.0008698**

Variants  
**1**

Interpretations  
**Likely consequential**

Disease Description

A carcinoma that arises from the lung and is characterized by the presence of malignant glandular epithelial cells. There is a male predilection with a male to female ratio of 2:1. Usually lung adenocarcinoma is asymptomatic and is identified through screening studies or as an incidental radiologic finding. If clinical symptoms are present they include shortness of breath, cough, hemoptysis, chest pain, and fever. Tobacco smoke is a known risk factor.

Gene Summary

| regulator of telomere elongation helicase 1

Known Drugs / Clinical Investigation

| No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.277 across 4 evidence rows and 4 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Uncommon

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|----------|-----------------|--------|--------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 20_63662850_G_A | missense_variant | Likely consequential    | 0.001287 | 0.0008698       | 0      | 0      | 25.2       | 0.6091            | 0.3284     | 0.519 | 0.3006   | 0.001287          | 0.0008698                  |

59. CYP2A6 - lung adenocarcinoma

ENSG00000255974 | EFO\_0000571

|                                    |                       |                                                          |
|------------------------------------|-----------------------|----------------------------------------------------------|
| L2G score<br>0.238                 | Unmet need index<br>5 | SAS global MAF<br>0.01779                                |
| Genome India global MAF<br>0.00764 | Variants<br>3         | Interpretations<br>Likely consequential;<br>not_missense |

Disease Description

A carcinoma that arises from the lung and is characterized by the presence of malignant glandular epithelial cells. There is a male predilection with a male to female ratio of 2:1. Usually lung adenocarcinoma is asymptomatic and is identified through screening studies or as an incidental radiologic finding. If clinical symptoms are present they include shortness of breath, cough, hemoptysis, chest pain, and fever. Tobacco smoke is a known risk factor.

Gene Summary

cytochrome P450 family 2 subfamily A member 6

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.238 across 3 evidence rows and 3 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Uncommon

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe   | AF_fin   | cadd_phred | am_pathogenicity | EVE_S CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|----------|-----------------|----------|----------|------------|------------------|------------|-------|----------|-------------------|----------------------------|
| 19_40843845_C_A | missense_variant | Likely consequential    | 0.001052 |                 | 0.000346 | 0.001268 | 23.1       | 0.6058           |            | 0.147 | 0.1383   | 0.01779           | 0.00764                    |
| 19_40843881_G_T | stop_gained      | not_missense            | 0.009121 |                 | 0.01919  | 0.003588 | 33         |                  |            |       |          | 0.01779           | 0.00764                    |
| 19_40848304_C_A | missense_variant | Likely consequential    | 0.007708 | 0.00764         | 0        | 0        | 29         | 0.7887           |            | 0.312 | 0.2805   | 0.01779           | 0.00764                    |

60. GPC5 - gastric cancer

ENSG00000179399 | MONDO\_0001056

|                                     |                       |                                         |
|-------------------------------------|-----------------------|-----------------------------------------|
| L2G score<br>0.869                  | Unmet need index<br>5 | SAS global MAF<br>0.004093              |
| Genome India global MAF<br>0.002456 | Variants<br>1         | Interpretations<br>Likely consequential |

Disease Description

A primary or metastatic malignant neoplasm involving the stomach.

Gene Summary

glypican 5

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: europepmc: score 0.712, evidence 2e+01; gwas\_credible\_sets: score 0.521, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.869, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|----------|-----------------|--------|--------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 13_91693637_C_G | missense_variant | Likely consequential    | 0.004093 | 0.002456        | 0      | 0      | 23.8       | 0.489            |           | 0.838 | 0.2864   | 0.004093          | 0.002456                   |



61. POU5F1B - gastric cancer

ENSG00000212993 | MONDO\_0001056

L2G score  
**0.828**

Unmet need index  
**5**

SAS global MAF  
**0.001041**

Genome India global MAF  
**0.002812**

Variants  
**2**

Interpretations  
**Likely consequential;  
not\_missense**

Disease Description

A primary or metastatic malignant neoplasm involving the stomach.

Gene Summary

POU class 5 homeobox 1B

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.671, evidence 3; europepmc: score 0.402, evidence 2e+01. GWAS credible-set evidence in this analysis has L2G 0.828, 3 evidence rows, and 3 study loci.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas    | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|-----------|-----------------|--------|--------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 8_127416359_G_C | missense_variant | Likely consequential    | 0.0008054 | 0.001586        | 0      | 0      | 20.9       | 0.6828           |           |       | 0.9479   | 0.001041          | 0.002812                   |
| 8_127416785_G_T | stop_gained      | not_missense            | 0.0002362 | 0.001228        | 0      | 0      | 35         |                  |           |       |          | 0.001041          | 0.002812                   |

62. HNF1B - gastric cancer

ENSG00000275410 | MONDO\_0001056

|                              |                       |                                 |
|------------------------------|-----------------------|---------------------------------|
| L2G score<br>0.71            | Unmet need index<br>5 | SAS global MAF<br>0.0169        |
| Genome India global MAF<br>0 | Variants<br>1         | Interpretations<br>not_missense |

Disease Description

A primary or metastatic malignant neoplasm involving the stomach.

Gene Summary

HNF1 homeobox B

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.494, evidence 2; europepmc: score 0.235, evidence 4. GWAS credible-set evidence in this analysis has L2G 0.71, 2 evidence rows, and 2 study loci.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid                  | consequence        | missense_interpretation | AF_sas | genome_india_af | AF_nfe | AF_fin  | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|------------------------|--------------------|-------------------------|--------|-----------------|--------|---------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 17_37710582_GTCACCAT_G | frameshift_variant | not_missense            | 0.0169 |                 | 0.0023 | 0.02255 | 35         |                  |           |       |          | 0.0169            | 0                          |

63. STPG2 - gastric cancer

ENSG00000163116 | MONDO\_0001056

|                                   |                       |                                         |
|-----------------------------------|-----------------------|-----------------------------------------|
| L2G score<br>0.342                | Unmet need index<br>5 | SAS global MAF<br>0.4415                |
| Genome India global MAF<br>0.4197 | Variants<br>1         | Interpretations<br>Likely consequential |

Disease Description

A primary or metastatic malignant neoplasm involving the stomach.

Gene Summary

| sperm tail PG-rich repeat containing 2

Known Drugs / Clinical Investigation

| No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.313, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.342, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid          | consequence      | missense_interpretation | AF_sas | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|----------------|------------------|-------------------------|--------|-----------------|--------|--------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 4_98128442_A_G | missense_variant | Likely consequential    | 0.4415 | 0.4197          | 0.4088 | 0.2902 | 24.6       | 0.6438           |           | 0.185 | 0.03216  | 0.4415            | 0.4197                     |

64. NR2E3 - gastric cancer

ENSG00000278570 | MONDO\_0001056

|                                     |                       |                                         |
|-------------------------------------|-----------------------|-----------------------------------------|
| L2G score<br>0.252                  | Unmet need index<br>5 | SAS global MAF<br>0.0007573             |
| Genome India global MAF<br>0.003121 | Variants<br>1         | Interpretations<br>Likely consequential |

Disease Description

A primary or metastatic malignant neoplasm involving the stomach.

Gene Summary

nuclear receptor subfamily 2 group E member 3

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.154, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.252, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid          | consequence      | missense_interpretation | AF_sas    | genome_india_af | AF_nfe    | AF_fin   | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|----------------|------------------|-------------------------|-----------|-----------------|-----------|----------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 15_7181966_G_A | missense_variant | Likely consequential    | 0.0007573 | 0.003121        | 0.0004058 | 0.006587 | 32         | 0.9546           | 0.6169    |       | 0.2446   | 0.0007573         | 0.003121                   |

65. L1TD1 - neuroblastoma

ENSG00000240563 | EFO\_0000621

|                                     |                          |                                 |
|-------------------------------------|--------------------------|---------------------------------|
| L2G score<br>0.931                  | Unmet need index<br>4.96 | SAS global MAF<br>0.004045      |
| Genome India global MAF<br>0.004042 | Variants<br>1            | Interpretations<br>not_missense |

Disease Description

Neuroblastoma (NB) is the most common solid, extracranial childhood tumor. It is an aggressive pediatric cancer that originates from neural crest tissues of the sympathetic nervous system.

Gene Summary

LINE1 type transposase domain containing 1

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.931 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid                  | consequence        | missense_interpretation | AF_sas   | genome_india_af | AF_nfe  | AF_fin  | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|------------------------|--------------------|-------------------------|----------|-----------------|---------|---------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 1_62210575_GA<br>ACT_G | frameshift_variant | not_missense            | 0.004045 | 0.004042        | 0.01217 | 0.00397 | 23.8       |                  |           |       |          | 0.004045          | 0.004042                   |

66. KANSL1L - neuroblastoma

ENSG00000144445 | EFO\_0000621

|                                     |                          |                                         |
|-------------------------------------|--------------------------|-----------------------------------------|
| L2G score<br>0.692                  | Unmet need index<br>4.96 | SAS global MAF<br>0.003556              |
| Genome India global MAF<br>0.004346 | Variants<br>2            | Interpretations<br>Likely consequential |

Disease Description

Neuroblastoma (NB) is the most common solid, extracranial childhood tumor. It is an aggressive pediatric cancer that originates from neural crest tissues of the sympathetic nervous system.

Gene Summary

KAT8 regulatory NSL complex subunit 1 like

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.692 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe   | AF_fin    | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|----------|-----------------|----------|-----------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 2_210028899_A_G | missense_variant | Likely consequential    | 0.001212 | 0.001177        | 0.001075 | 0.0002624 | 25.1       | 0.8966           |           | 0.437 | 0.04716  | 0.003556          | 0.004346                   |
| 2_210043964_G_C | missense_variant | Likely consequential    | 0.002347 | 0.003173        | 0        | 0         | 22.1       | 0.8495           |           | 0.196 | 0.2891   | 0.003556          | 0.004346                   |

67. WRN - pneumococcal pneumonia

ENSG00000165392 | EFO\_1001474

|                                    |                          |                                 |
|------------------------------------|--------------------------|---------------------------------|
| L2G score<br>0.585                 | Unmet need index<br>4.94 | SAS global MAF<br>0.01562       |
| Genome India global MAF<br>0.01724 | Variants<br>1            | Interpretations<br>not_missense |

Disease Description

A febrile disease caused by STREPTOCOCCUS PNEUMONIAE.

Gene Summary

WRN RecQ like helicase

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.318, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.585, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Uncommon

Alleles and Scores

| varid          | consequence | missense_interpretation | AF_sas  | genome_india_af | AF_nfe   | AF_fin   | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|----------------|-------------|-------------------------|---------|-----------------|----------|----------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 8_31173019_C_T | stop_gained | not_missense            | 0.01562 | 0.01724         | 0.001536 | 0.001143 | 35         |                  |           |       |          | 0.01562           | 0.01724                    |

68. ASB5 - pneumococcal pneumonia

ENSG00000164122 | EFO\_1001474

|                                     |                          |                                         |
|-------------------------------------|--------------------------|-----------------------------------------|
| L2G score<br>0.153                  | Unmet need index<br>4.94 | SAS global MAF<br>0.0001748             |
| Genome India global MAF<br>0.001023 | Variants<br>1            | Interpretations<br>Likely consequential |

Disease Description

A febrile disease caused by STREPTOCOCCUS PNEUMONIAE.

Gene Summary

ankyrin repeat and SOCS box containing 5

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.115, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.153, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Uncommon

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas    | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|-----------|-----------------|--------|--------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 4_176215715_G_A | missense_variant | Likely consequential    | 0.0001748 | 0.001023        | 0      | 0      | 25         | 0.3069           |           | 0.378 | 0.9872   | 0.0001748         | 0.001023                   |



69. ACKR1 - microscopic polyangiitis

ENSG00000213088 | EFO\_1000784

|                                   |                          |                                         |
|-----------------------------------|--------------------------|-----------------------------------------|
| L2G score<br>0.889                | Unmet need index<br>4.91 | SAS global MAF<br>0.0004407             |
| Genome India global MAF<br>0.0044 | Variants<br>1            | Interpretations<br>Likely consequential |

Disease Description

Microscopic polyangiitis (MPA) is an inflammatory, necrotizing, systemic vasculitis that affects predominantly small vessels (i.e. small arteries, arterioles, capillaries, venules) in multiple organs.

Gene Summary

atypical chemokine receptor 1 (Duffy blood group)

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.889 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas    | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|-----------|-----------------|--------|--------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 1_159205834_G_A | missense_variant | Likely consequential    | 0.0004407 | 0.0044          | 0      | 0      | 21.3       | 0.8225           | 0.579     | 0.143 | 0.0569   | 0.0004407         | 0.0044                     |

70. ADH1B - oropharynx cancer

ENSG00000196616 | EFO\_1001931

|                                     |                          |                                         |
|-------------------------------------|--------------------------|-----------------------------------------|
| L2G score<br>0.965                  | Unmet need index<br>4.86 | SAS global MAF<br>0.00622               |
| Genome India global MAF<br>0.003426 | Variants<br>2            | Interpretations<br>Likely consequential |

Disease Description

A primary or metastatic malignant neoplasm that affects the oropharynx.

Gene Summary

alcohol dehydrogenase 1B (class I), beta polypeptide

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.965 across 2 evidence rows and 2 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid          | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe   | AF_fin    | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|----------------|------------------|-------------------------|----------|-----------------|----------|-----------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 4_99307859_C_T | missense_variant | Likely consequential    | 0.004835 | 0.002456        | 0.003268 | 0.0005805 | 24.1       | 0.7715           |           | 0.218 | 0.1022   | 0.00622           | 0.003426                   |
| 4_99307860_G_A | missense_variant | Likely consequential    | 0.001391 | 0.0009722       | 0.001129 | 5.617e-05 | 27.2       | 0.7617           |           | 0.194 | 0.1033   | 0.00622           | 0.003426                   |

71. INPP5D - oropharynx cancer

ENSG00000168918 | EFO\_1001931

|                                     |                          |                                         |
|-------------------------------------|--------------------------|-----------------------------------------|
| L2G score<br>0.745                  | Unmet need index<br>4.86 | SAS global MAF<br>0.002724              |
| Genome India global MAF<br>0.001689 | Variants<br>1            | Interpretations<br>Likely consequential |

Disease Description

A primary or metastatic malignant neoplasm that affects the oropharynx.

Gene Summary

inositol polyphosphate-5-phosphatase D

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.745 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|----------|-----------------|--------|--------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 2_233158415_C_T | missense_variant | Likely consequential    | 0.002724 | 0.001689        | 0      | 0      | 25.1       | 0.8393           |           |       | 0.312    | 0.002724          | 0.001689                   |

## 72. ANKS1A - oropharynx cancer

ENSG00000064999 | EFO\_1001931

|                                            |                                 |                                                |
|--------------------------------------------|---------------------------------|------------------------------------------------|
| L2G score<br><b>0.612</b>                  | Unmet need index<br><b>4.86</b> | SAS global MAF<br><b>0.002968</b>              |
| Genome India global MAF<br><b>0.001075</b> | Variants<br><b>1</b>            | Interpretations<br><b>Likely consequential</b> |

### Disease Description

A primary or metastatic malignant neoplasm that affects the oropharynx.

### Gene Summary

| ankyrin repeat and sterile alpha motif domain containing 1A

### Known Drugs / Clinical Investigation

| No same target-disease clinical indication found in the downloaded Open Targets drug tables.

### Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.612 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

### SAS / Indian Prevalence

Rare

### Alleles and Scores

| varid          | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe   | AF_fin    | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|----------------|------------------|-------------------------|----------|-----------------|----------|-----------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 6_35078619_G_A | missense_variant | Likely consequential    | 0.002968 | 0.001075        | 0.001008 | 0.0001319 | 29.4       | 0.1623           |           | 0.785 | 0.1089   | 0.002968          | 0.001075                   |

### 73. CAPN13 - oropharynx cancer

ENSG00000162949 | EFO\_1001931

L2G score  
**0.502**

Unmet need index  
**4.86**

SAS global MAF  
**0.003337**

Genome India global MAF  
**0.008084**

Variants  
**1**

Interpretations  
**Likely consequential**

#### Disease Description

A primary or metastatic malignant neoplasm that affects the oropharynx.

#### Gene Summary

calpain 13

#### Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

#### Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.502 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

#### SAS / Indian Prevalence

Rare

#### Alleles and Scores

| varid          | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe   | AF_fin  | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|----------------|------------------|-------------------------|----------|-----------------|----------|---------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 2_30763111_G_A | missense_variant | Likely consequential    | 0.003337 | 0.008084        | 0.005711 | 0.01445 | 26.2       | 0.5726           |           | 0.62  | 0.04429  | 0.003337          | 0.008084                   |

74. SULT1C4 - oropharynx cancer

ENSG00000198075 | EFO\_1001931

|                                      |                          |                                 |
|--------------------------------------|--------------------------|---------------------------------|
| L2G score<br>0.353                   | Unmet need index<br>4.86 | SAS global MAF<br>0.001521      |
| Genome India global MAF<br>0.0009722 | Variants<br>1            | Interpretations<br>not_missense |

Disease Description

A primary or metastatic malignant neoplasm that affects the oropharynx.

Gene Summary

| sulfotransferase family 1C member 4

Known Drugs / Clinical Investigation

| No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.353 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid           | consequence          | missense_interpretation | AF_sas   | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|----------------------|-------------------------|----------|-----------------|--------|--------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 2_108383512_T_C | splice_donor_variant | not_missense            | 0.001521 | 0.0009722       | 0      | 0      | 24.7       |                  |           |       |          | 0.001521          | 0.0009722                  |

75. GPR27 - oropharynx cancer

ENSG00000170837 | EFO\_1001931

L2G score  
**0.197**

Unmet need index  
**4.86**

SAS global MAF  
**0.003987**

Genome India global MAF  
**0**

Variants  
**2**

Interpretations  
**Likely consequential**

Disease Description

A primary or metastatic malignant neoplasm that affects the oropharynx.

Gene Summary

G protein-coupled receptor 27

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.197 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid          | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe   | AF_fin    | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|----------------|------------------|-------------------------|----------|-----------------|----------|-----------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 3_71755049_A_C | missense_variant | Likely consequential    | 0.001837 |                 | 0.006974 | 0.0002525 | 26         | 0.7877           |           | 0.876 | 0.9995   | 0.003987          | 0                          |
| 3_71755050_A_C | missense_variant | Likely consequential    | 0.002153 |                 | 0.007921 | 0.0002732 | 26.5       | 0.7119           |           | 0.888 | 0.9984   | 0.003987          | 0                          |

76. RTL1 - oropharynx cancer

ENSG00000254656 | EFO\_1001931

|                                     |                          |                                         |
|-------------------------------------|--------------------------|-----------------------------------------|
| L2G score<br>0.168                  | Unmet need index<br>4.86 | SAS global MAF<br>0.0008117             |
| Genome India global MAF<br>0.001893 | Variants<br>1            | Interpretations<br>Likely consequential |

Disease Description

A primary or metastatic malignant neoplasm that affects the oropharynx.

Gene Summary

retrotransposon Gag like 1

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.168 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid            | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|------------------|------------------|-------------------------|----------|-----------------|--------|--------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 14_100884608_C_T | missense_variant | Likely consequential    | 0.000817 | 0.001893        | 0      | 0      | 23.1       | 0.092            |           |       | 0.9187   | 0.000817          | 0.001893                   |



77. DPY19L4 - polyarteritis nodosa

ENSG00000156162 | MONDO\_0019170

L2G score  
0.73

Unmet need index  
4.85

SAS global MAF  
0.01152

Genome India global MAF  
0.002149

Variants  
4

Interpretations  
Likely consequential;  
not\_missense

Disease Description

Polyarteritis nodosa (PAN) is a rare, clinically heterogeneous, rheumatologic disease characterized by necrotizing inflammatory lesions affecting small- and medium-sized blood vessels. PAN most commonly affects skin, joints, peripheral nerves, the gut, and the kidney.

Gene Summary

dpy-19 like 4

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.373, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.73, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid          | consequence             | missense_interpretation | AF_sas   | genome_india_af | AF_nfe    | AF_fin   | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|----------------|-------------------------|-------------------------|----------|-----------------|-----------|----------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 8_94734629_G_A | splice_acceptor_variant | not_misense             | 0.001467 | 0.002149        | 0         | 0        | 32         |                   |            |       |          | 0.01152           | 0.002149                   |
| 8_94781082_A_T | splice_acceptor_variant | not_misense             | 0.002889 |                 | 0.000628  | 0.003751 | 35         |                   |            |       |          | 0.01152           | 0.002149                   |
| 8_94781083_G_T | splice_acceptor_variant | not_misense             | 0.004793 |                 | 0.01267   | 0.003047 | 35         |                   |            |       |          | 0.01152           | 0.002149                   |
| 8_94789792_G_A | missense_variant        | Likely consequential    | 0.00242  |                 | 4.232e-05 | 0.002081 | 26.4       | 0.5924            |            | 0.253 | 0.2076   | 0.01152           | 0.002149                   |

78. NEDD1 - polyarteritis nodosa

ENSG00000139350 | MONDO\_0019170

L2G score  
**0.504**

Unmet need index  
**4.85**

SAS global MAF  
**2.372e-05**

Genome India global MAF  
**0.001228**

Variants  
**1**

Interpretations  
**Likely consequential**

Disease Description

Polyarteritis nodosa (PAN) is a rare, clinically heterogeneous, rheumatologic disease characterized by necrotizing inflammatory lesions affecting small- and medium-sized blood vessels. PAN most commonly affects skin, joints, peripheral nerves, the gut, and the kidney.

Gene Summary

NEDD1 gamma-tubulin ring complex targeting factor

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.396, evidence 2. GWAS credible-set evidence in this analysis has L2G 0.504, 2 evidence rows, and 2 study loci.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas    | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|-----------|-----------------|--------|--------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 12_96937212_A_C | missense_variant | Likely consequential    | 2.372e-05 | 0.001228        | 0      | 0      | 21         | 0.3107            |            | 0.146 | 0.7277   | 2.372e-05         | 0.001228                   |

79. GCFC2 - polyarteritis nodosa

ENSG00000005436 | MONDO\_0019170

L2G score  
**0.501**

Unmet need index  
**4.85**

SAS global MAF  
**0.000535**

Genome India global MAF  
**0.001126**

Variants  
**1**

Interpretations  
**not\_missense**

Disease Description

Polyarteritis nodosa (PAN) is a rare, clinically heterogeneous, rheumatologic disease characterized by necrotizing inflammatory lesions affecting small- and medium-sized blood vessels. PAN most commonly affects skin, joints, peripheral nerves, the gut, and the kidney.

Gene Summary

GC-rich sequence DNA-binding factor 2

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.334, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.501, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid           | consequence        | missense_interpretation | AF_sas   | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_S CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|--------------------|-------------------------|----------|-----------------|--------|--------|------------|------------------|------------|-------|----------|-------------------|----------------------------|
| 2_75692013_AT_A | frameshift_variant | not_missense            | 0.000535 | 0.001126        | 0      | 0      | 24.4       |                  |            |       |          | 0.000535          | 0.001126                   |

80. SEMA3C - polyarteritis nodosa

ENSG00000075223 | MONDO\_0019170

|                                    |                          |                                         |
|------------------------------------|--------------------------|-----------------------------------------|
| L2G score<br>0.48                  | Unmet need index<br>4.85 | SAS global MAF<br>0.03783               |
| Genome India global MAF<br>0.05051 | Variants<br>1            | Interpretations<br>Likely consequential |

Disease Description

Polyarteritis nodosa (PAN) is a rare, clinically heterogeneous, rheumatologic disease characterized by necrotizing inflammatory lesions affecting small- and medium-sized blood vessels. PAN most commonly affects skin, joints, peripheral nerves, the gut, and the kidney.

Gene Summary

semaphorin 3C

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.393, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.48, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid          | consequence      | missense_interpretation | AF_sas  | genome_india_af | AF_nfe   | AF_fin   | cadd_phred | am_pathogenicity | EVE_S CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|----------------|------------------|-------------------------|---------|-----------------|----------|----------|------------|------------------|------------|-------|----------|-------------------|----------------------------|
| 7_80798214_C_T | missense_variant | Likely consequential    | 0.03783 | 0.05051         | 0.006466 | 0.001262 | 32         | 0.7772           |            | 0.353 | 0.08897  | 0.03783           | 0.05051                    |

81. C7orf33 - polyarteritis nodosa

ENSG00000170279 | MONDO\_0019170

|                                    |                          |                                 |
|------------------------------------|--------------------------|---------------------------------|
| L2G score<br>0.392                 | Unmet need index<br>4.85 | SAS global MAF<br>0.05077       |
| Genome India global MAF<br>0.04226 | Variants<br>1            | Interpretations<br>not_missense |

Disease Description

Polyarteritis nodosa (PAN) is a rare, clinically heterogeneous, rheumatologic disease characterized by necrotizing inflammatory lesions affecting small- and medium-sized blood vessels. PAN most commonly affects skin, joints, peripheral nerves, the gut, and the kidney.

Gene Summary

chromosome 7 open reading frame 33

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.165, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.392, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid           | consequence | missense_interpretation | AF_sas  | genome_india_af | AF_nfe  | AF_fin  | cadd_phred | am_pathogenicity | EVE_S CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|-------------|-------------------------|---------|-----------------|---------|---------|------------|------------------|------------|-------|----------|-------------------|----------------------------|
| 7_148590928_G_T | start_loss  | not_missense            | 0.05077 | 0.04226         | 0.03583 | 0.01586 | 7.765      |                  |            |       | 0.01417  | 0.05077           | 0.04226                    |

82. ELFN1 - polyarteritis nodosa

ENSG00000225968 | MONDO\_0019170

|                              |                          |                                                          |
|------------------------------|--------------------------|----------------------------------------------------------|
| L2G score<br>0.222           | Unmet need index<br>4.85 | SAS global MAF<br>0.05891                                |
| Genome India global MAF<br>0 | Variants<br>4            | Interpretations<br>Likely consequential;<br>not_missense |

Disease Description

Polyarteritis nodosa (PAN) is a rare, clinically heterogeneous, rheumatologic disease characterized by necrotizing inflammatory lesions affecting small- and medium-sized blood vessels. PAN most commonly affects skin, joints, peripheral nerves, the gut, and the kidney.

Gene Summary

extracellular leucine rich repeat and fibronectin type III domain containing 1

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.0987, evidence 1; europepmc: score 0.0304, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.222, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid                      | consequence        | missense_interpretation | AF_sas   | genome_india_af | AF_nfe   | AF_fin   | cadd_phred | am_pathogenicity | EVE_S CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|----------------------------|--------------------|-------------------------|----------|-----------------|----------|----------|------------|------------------|------------|-------|----------|-------------------|----------------------------|
| 7_1746231_C_CTTTTTAAATGATA | frameshift_variant | not_misense             | 0.006063 |                 | 0.001199 | 0.000262 | 23.1       |                  |            |       |          | 0.05891           | 0                          |
| 7_1746232_C_CG             | frameshift_variant | not_misense             | 0.01307  |                 | 0.00324  | 0.001124 | 24.3       |                  |            |       |          | 0.05891           | 0                          |
| 7_1746242_AGAGTTCGGTGG_A   | frameshift_variant | not_misense             | 0.01534  |                 | 0.004613 | 0.00178  | 34         |                  |            |       |          | 0.05891           | 0                          |
| 7_1746263_C_A              | missense_variant   | Likely consequential    | 0.02569  |                 | 0.006784 | 0.003496 | 24.4       | 0.7519           |            |       |          | 0.05891           | 0                          |

83. TYW1B - polyarteritis nodosa

ENSG00000277149 | MONDO\_0019170

L2G score  
**0.164**

Unmet need index  
**4.85**

SAS global MAF  
**0.002548**

Genome India global MAF  
**0**

Variants  
**1**

Interpretations  
**not\_missense**

Disease Description

Polyarteritis nodosa (PAN) is a rare, clinically heterogeneous, rheumatologic disease characterized by necrotizing inflammatory lesions affecting small- and medium-sized blood vessels. PAN most commonly affects skin, joints, peripheral nerves, the gut, and the kidney.

Gene Summary

| tRNA-yW synthesizing protein 1 homolog B

Known Drugs / Clinical Investigation

| No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.105, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.164, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid          | consequence | missense_interpretation | AF_sas   | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_S CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|----------------|-------------|-------------------------|----------|-----------------|--------|--------|------------|------------------|------------|-------|----------|-------------------|----------------------------|
| 7_72575498_T_C | stop_loss   | not_missense            | 0.002548 |                 | 0      | 0      | 17.83      |                  |            |       |          | 0.002548          | 0                          |

84. BLK - esophageal adenocarcinoma

ENSG00000136573 | EFO\_0000478

L2G score  
**0.831**

Unmet need index  
**4.84**

SAS global MAF  
**0.005038**

Genome India global MAF  
**0.002659**

Variants  
**2**

Interpretations  
**Likely consequential**

Disease Description

A malignant tumor with glandular differentiation arising predominantly from Barrett mucosa in the lower third of the esophagus. Rare examples of esophageal adenocarcinoma deriving from ectopic gastric mucosa in the upper esophagus have also been reported. Grossly, esophageal adenocarcinomas are similar to esophageal squamous cell carcinomas. Microscopically, adenocarcinomas arising in the setting of Barrett esophagus are typically papillary and/or tubular. The prognosis is poor.

Gene Summary

BLK proto-oncogene, Src family tyrosine kinase

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.831 across 3 evidence rows and 3 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid          | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe   | AF_fin    | cadd_phred | am_pathogenicity | EVE_S CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|----------------|------------------|-------------------------|----------|-----------------|----------|-----------|------------|------------------|------------|-------|----------|-------------------|----------------------------|
| 8_11557983_A_C | missense_variant | Likely consequential    | 0.002864 | 0.0008187       | 0.002022 | 9.361e-05 | 26.1       | 0.5985           |            | 0.829 | 0.06809  | 0.005038          | 0.002659                   |
| 8_11561347_C_T | missense_variant | Likely consequential    | 0.002181 | 0.001842        | 0.001095 | 0.0005805 | 33         | 0.8702           |            | 0.823 | 0.1829   | 0.005038          | 0.002659                   |



85. NNT - esophageal adenocarcinoma

ENSG00000112992 | EFO\_0000478

|                                    |                          |                                         |
|------------------------------------|--------------------------|-----------------------------------------|
| L2G score<br>0.325                 | Unmet need index<br>4.84 | SAS global MAF<br>0.02007               |
| Genome India global MAF<br>0.01775 | Variants<br>1            | Interpretations<br>Likely consequential |

Disease Description

A malignant tumor with glandular differentiation arising predominantly from Barrett mucosa in the lower third of the esophagus. Rare examples of esophageal adenocarcinoma deriving from ectopic gastric mucosa in the upper esophagus have also been reported. Grossly, esophageal adenocarcinomas are similar to esophageal squamous cell carcinomas. Microscopically, adenocarcinomas arising in the setting of Barrett esophagus are typically papillary and/or tubular. The prognosis is poor.

Gene Summary

nicotinamide nucleotide transhydrogenase

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.325 across 1 evidence rows and 1 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid          | consequence      | missense_interpretation | AF_sas  | genome_india_af | AF_nfe   | AF_fin  | cadd_phred | am_pathogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|----------------|------------------|-------------------------|---------|-----------------|----------|---------|------------|------------------|-----------|-------|----------|-------------------|----------------------------|
| 5_43700219_A_G | missense_variant | Likely consequential    | 0.02007 | 0.01775         | 0.004595 | 0.01125 | 23.5       | 0.1974           |           | 0.691 | 0.02754  | 0.02007           | 0.01775                    |

86. UST - amyotrophic lateral sclerosis

ENSG00000111962 | MONDO\_0004976

L2G score  
**0.771**

Unmet need index  
**4.81**

SAS global MAF  
**2.319e-05**

Genome India global MAF  
**0.9999**

Variants  
**1**

Interpretations  
**not\_missense**

Disease Description

Amyotrophic lateral sclerosis (ALS) is a neurodegenerative disease characterized by progressive muscular paralysis reflecting degeneration of motor neurons in the primary motor cortex, corticospinal tracts, brainstem and spinal cord.

Gene Summary

uronyl 2-sulfotransferase

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.502, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.771, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid            | consequence        | missense_interpretation | AF_sas    | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_S CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|------------------|--------------------|-------------------------|-----------|-----------------|--------|--------|------------|------------------|------------|-------|----------|-------------------|----------------------------|
| 6_149074005_GC_G | frameshift_variant | not_missense            | 2.319e-05 | 0.9999          | 0      | 0      |            |                  |            |       |          | 2.319e-05         | 0.9999                     |

87. DUSP16 - amyotrophic lateral sclerosis

ENSG00000111266 | MONDO\_0004976

L2G score  
0.751

Unmet need index  
4.81

SAS global MAF  
0.2318

Genome India global MAF  
0.1593

Variants  
2

Interpretations  
not\_missense

Disease Description

Amyotrophic lateral sclerosis (ALS) is a neurodegenerative disease characterized by progressive muscular paralysis reflecting degeneration of motor neurons in the primary motor cortex, corticospinal tracts, brainstem and spinal cord.

Gene Summary

dual specificity phosphatase 16

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.416, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.751, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid                | consequence        | missense_interpretation | AF_sas | genome_india_af | AF_nfe   | AF_fin  | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|----------------------|--------------------|-------------------------|--------|-----------------|----------|---------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 12_12477741_GCACGC_G | frameshift_variant | not_missense            | 0.1236 | 0.1593          | 0.00336  | 0.02682 | 23.4       |                   |            |       |          | 0.2318            | 0.1593                     |
| 12_12477747_TGGGC_T  | frameshift_variant | not_missense            | 0.1235 |                 | 0.003357 | 0.02677 | 26.9       |                   |            |       |          | 0.2318            | 0.1593                     |

88. ANKS1B - amyotrophic lateral sclerosis

ENSG00000185046 | MONDO\_0004976

|                                     |                          |                                         |
|-------------------------------------|--------------------------|-----------------------------------------|
| L2G score<br>0.75                   | Unmet need index<br>4.81 | SAS global MAF<br>8.115e-05             |
| Genome India global MAF<br>0.001228 | Variants<br>1            | Interpretations<br>Likely consequential |

Disease Description

Amyotrophic lateral sclerosis (ALS) is a neurodegenerative disease characterized by progressive muscular paralysis reflecting degeneration of motor neurons in the primary motor cortex, corticospinal tracts, brainstem and spinal cord.

Gene Summary

ankyrin repeat and sterile alpha motif domain containing 1B

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.391, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.75, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas    | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|-----------|-----------------|--------|--------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 12_99806627_T_C | missense_variant | Likely consequential    | 8.115e-05 | 0.001228        | 0      | 0      | 27.7       | 0.2076            |            |       | 0.7108   | 8.115e-05         | 0.001228                   |

89. ACOXL - amyotrophic lateral sclerosis

ENSG00000153093 | MONDO\_0004976

|                                     |                          |                                         |
|-------------------------------------|--------------------------|-----------------------------------------|
| L2G score<br>0.674                  | Unmet need index<br>4.81 | SAS global MAF<br>0.0007883             |
| Genome India global MAF<br>0.001842 | Variants<br>1            | Interpretations<br>Likely consequential |

Disease Description

Amyotrophic lateral sclerosis (ALS) is a neurodegenerative disease characterized by progressive muscular paralysis reflecting degeneration of motor neurons in the primary motor cortex, corticospinal tracts, brainstem and spinal cord.

Gene Summary

acyl-CoA oxidase like

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.41, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.674, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas    | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|-----------|-----------------|--------|--------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 2_110794129_C_G | missense_variant | Likely consequential    | 0.0007883 | 0.001842        | 0      | 0      | 6.057      | 0.8466            |            | 0.601 | 0.8359   | 0.0007883         | 0.001842                   |

90. G2E3 - amyotrophic lateral sclerosis

ENSG00000092140 | MONDO\_0004976

|                              |                          |                                 |
|------------------------------|--------------------------|---------------------------------|
| L2G score<br>0.643           | Unmet need index<br>4.81 | SAS global MAF<br>0.02063       |
| Genome India global MAF<br>0 | Variants<br>1            | Interpretations<br>not_missense |

Disease Description

Amyotrophic lateral sclerosis (ALS) is a neurodegenerative disease characterized by progressive muscular paralysis reflecting degeneration of motor neurons in the primary motor cortex, corticospinal tracts, brainstem and spinal cord.

Gene Summary

G2/M-phase specific E3 ubiquitin protein ligase

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.415, evidence 2; europepmc: score 0.27, evidence 1e+01. GWAS credible-set evidence in this analysis has L2G 0.643, 2 evidence rows, and 2 study loci.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid            | consequence        | missense_interpretation | AF_sas  | genome_india_af | AF_nfe  | AF_fin   | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|------------------|--------------------|-------------------------|---------|-----------------|---------|----------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 14_30605565_T_TA | frameshift_variant | not_missense            | 0.02063 |                 | 0.02155 | 0.008025 | 27.3       |                   |            |       |          | 0.02063           | 0                          |

91. CRYZ - amyotrophic lateral sclerosis

ENSG00000116791 | MONDO\_0004976

|                              |                          |                                 |
|------------------------------|--------------------------|---------------------------------|
| L2G score<br>0.608           | Unmet need index<br>4.81 | SAS global MAF<br>0.001601      |
| Genome India global MAF<br>0 | Variants<br>1            | Interpretations<br>not_missense |

Disease Description

Amyotrophic lateral sclerosis (ALS) is a neurodegenerative disease characterized by progressive muscular paralysis reflecting degeneration of motor neurons in the primary motor cortex, corticospinal tracts, brainstem and spinal cord.

Gene Summary

crystallin zeta

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.353, evidence 1; europepmc: score 0.23, evidence 3. GWAS credible-set evidence in this analysis has L2G 0.608, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid              | consequence        | missense_interpretation | AF_sas   | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|--------------------|--------------------|-------------------------|----------|-----------------|--------|--------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 1_74706961_GGTTT_G | frameshift_variant | not_missense            | 0.001601 |                 | 0      | 0      | 29.7       |                   |            |       |          | 0.001601          | 0                          |

92. NFIA - amyotrophic lateral sclerosis

ENSG00000162599 | MONDO\_0004976

|                              |                          |                                 |
|------------------------------|--------------------------|---------------------------------|
| L2G score<br>0.52            | Unmet need index<br>4.81 | SAS global MAF<br>0.1045        |
| Genome India global MAF<br>0 | Variants<br>3            | Interpretations<br>not_missense |

Disease Description

Amyotrophic lateral sclerosis (ALS) is a neurodegenerative disease characterized by progressive muscular paralysis reflecting degeneration of motor neurons in the primary motor cortex, corticospinal tracts, brainstem and spinal cord.

Gene Summary

nuclear factor I A

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.397, evidence 1; impc: score 0.324, evidence 2. GWAS credible-set evidence in this analysis has L2G 0.52, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid            | consequence             | missense_interpretation | AF_sas   | genome_india_af | AF_nfe   | AF_fin   | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|------------------|-------------------------|-------------------------|----------|-----------------|----------|----------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 1_61406559_CAG_C | splice_acceptor_variant | not_missense            | 0.007711 |                 | 0.001255 | 0.004074 | 34         |                   |            |       |          | 0.1045            | 0                          |
| 1_61406560_A_C   | splice_acceptor_variant | not_missense            | 0.06294  |                 | 0.01306  | 0.02819  | 35         |                   |            |       |          | 0.1045            | 0                          |
| 1_61406561_G_C   | splice_acceptor_variant | not_missense            | 0.03689  |                 | 0.009639 | 0.01882  | 36         |                   |            |       |          | 0.1045            | 0                          |



93. WDR72 - amyotrophic lateral sclerosis

ENSG00000166415 | MONDO\_0004976

L2G score  
**0.459**

Unmet need index  
**4.81**

SAS global MAF  
**0.2429**

Genome India global MAF  
**0.259**

Variants  
**1**

Interpretations  
**Likely consequential**

Disease Description

Amyotrophic lateral sclerosis (ALS) is a neurodegenerative disease characterized by progressive muscular paralysis reflecting degeneration of motor neurons in the primary motor cortex, corticospinal tracts, brainstem and spinal cord.

Gene Summary

WD repeat domain 72

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.31, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.459, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|--------|-----------------|--------|--------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 15_53615751_G_A | missense_variant | Likely consequential    | 0.2429 | 0.259           | 0.2251 | 0.257  | 25.2       | 0.3382            | 0.7526     | 0.528 | 0.02449  | 0.2429            | 0.259                      |

94. NDST4 - amyotrophic lateral sclerosis

ENSG00000138653 | MONDO\_0004976

|                                     |                          |                                         |
|-------------------------------------|--------------------------|-----------------------------------------|
| L2G score<br>0.451                  | Unmet need index<br>4.81 | SAS global MAF<br>0.001049              |
| Genome India global MAF<br>0.003223 | Variants<br>1            | Interpretations<br>Likely consequential |

Disease Description

Amyotrophic lateral sclerosis (ALS) is a neurodegenerative disease characterized by progressive muscular paralysis reflecting degeneration of motor neurons in the primary motor cortex, corticospinal tracts, brainstem and spinal cord.

Gene Summary

N-deacetylase and N-sulfotransferase 4

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.346, evidence 1; europepmc: score 0.259, evidence 2. GWAS credible-set evidence in this analysis has L2G 0.451, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|----------|-----------------|--------|--------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 4_114829860_A_G | missense_variant | Likely consequential    | 0.001049 | 0.003223        | 0      | 0      | 25.2       | 0.4293            |            | 0.555 | 0.169    | 0.001049          | 0.003223                   |

95. CCNA1 - amyotrophic lateral sclerosis

ENSG00000133101 | MONDO\_0004976

|                              |                          |                                         |
|------------------------------|--------------------------|-----------------------------------------|
| L2G score<br>0.367           | Unmet need index<br>4.81 | SAS global MAF<br>0.003142              |
| Genome India global MAF<br>0 | Variants<br>2            | Interpretations<br>Likely consequential |

Disease Description

Amyotrophic lateral sclerosis (ALS) is a neurodegenerative disease characterized by progressive muscular paralysis reflecting degeneration of motor neurons in the primary motor cortex, corticospinal tracts, brainstem and spinal cord.

Gene Summary

cyclin A1

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.189, evidence 1; europepmc: score 0.0122, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.367, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe   | AF_fin   | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|----------|-----------------|----------|----------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 13_36438732_A_G | missense_variant | Likely consequential    | 0.002022 |                 | 0.001043 | 0.01569  | 30         |                   |            | 0.682 | 0.1137   | 0.003142          | 0                          |
| 13_36438735_T_G | missense_variant | Likely consequential    | 0.001122 |                 | 0.000366 | 0.002806 | 27.9       |                   |            | 0.739 | 0.08409  | 0.003142          | 0                          |

96. KIAA0232 - amyotrophic lateral sclerosis

ENSG00000170871 | MONDO\_0004976

L2G score  
0.311

Unmet need index  
4.81

SAS global MAF  
0.001461

Genome India global MAF  
0.002149

Variants  
1

Interpretations  
Likely consequential

Disease Description

Amyotrophic lateral sclerosis (ALS) is a neurodegenerative disease characterized by progressive muscular paralysis reflecting degeneration of motor neurons in the primary motor cortex, corticospinal tracts, brainstem and spinal cord.

Gene Summary

KIAA0232

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.267, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.311, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid         | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|---------------|------------------|-------------------------|----------|-----------------|--------|--------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 4_6863249_G_A | missense_variant | Likely consequential    | 0.001461 | 0.002149        | 0      | 0      | 21.2       | 0.2358            |            | 0.527 | 0.105    | 0.001461          | 0.002149                   |

97. MTNR1B - amyotrophic lateral sclerosis

ENSG00000134640 | MONDO\_0004976

|                                     |                          |                                         |
|-------------------------------------|--------------------------|-----------------------------------------|
| L2G score<br>0.305                  | Unmet need index<br>4.81 | SAS global MAF<br>0.001924              |
| Genome India global MAF<br>0.001842 | Variants<br>1            | Interpretations<br>Likely consequential |

Disease Description

Amyotrophic lateral sclerosis (ALS) is a neurodegenerative disease characterized by progressive muscular paralysis reflecting degeneration of motor neurons in the primary motor cortex, corticospinal tracts, brainstem and spinal cord.

Gene Summary

melatonin receptor 1B

Known Drugs / Clinical Investigation

MELATONIN (PHASE\_2; Melatonin receptor agonist)

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.23, evidence 1; clinical\_precedence: score 0.122, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.305, 1 evidence rows, and 1 study loci.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas   | genome_india_af | AF_nfe   | AF_fin    | cadd_phred | am_pat_hogenicity | EVE_SCORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|----------|-----------------|----------|-----------|------------|-------------------|-----------|-------|----------|-------------------|----------------------------|
| 11_92981635_C_T | missense_variant | Likely consequential    | 0.001924 | 0.001842        | 0.001504 | 5.616e-05 | 18.72      | 0.7634            |           | 0.709 | 0.1385   | 0.001924          | 0.001842                   |

98. GGT5 - amyotrophic lateral sclerosis

ENSG00000099998 | MONDO\_0004976

|                              |                          |                                                          |
|------------------------------|--------------------------|----------------------------------------------------------|
| L2G score<br>0.169           | Unmet need index<br>4.81 | SAS global MAF<br>0.05402                                |
| Genome India global MAF<br>0 | Variants<br>2            | Interpretations<br>Likely consequential;<br>not_missense |

Disease Description

Amyotrophic lateral sclerosis (ALS) is a neurodegenerative disease characterized by progressive muscular paralysis reflecting degeneration of motor neurons in the primary motor cortex, corticospinal tracts, brainstem and spinal cord.

Gene Summary

gamma-glutamyltransferase 5

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

Datasource-level direct associations: gwas\_credible\_sets: score 0.164, evidence 2; europepmc: score 0.0243, evidence 1. GWAS credible-set evidence in this analysis has L2G 0.169, 2 evidence rows, and 2 study loci.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid           | consequence             | missense_interpretation | AF_sas   | genome_india_af | AF_nfe    | AF_fin   | cadd_phred | am_pathogenicity | EVE_S CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|-------------------------|-------------------------|----------|-----------------|-----------|----------|------------|------------------|------------|-------|----------|-------------------|----------------------------|
| 22_24232203_A_G | missense_variant        | Likely consequential    | 0.003823 |                 | 0.0009495 | 0.001735 | 27.7       | 0.6413           |            | 0.403 | 0.381    | 0.05402           | 0                          |
| 22_24232210_T_G | splice_acceptor_variant | not_misense             | 0.05039  |                 | 0.02001   | 0.01907  | 29.6       |                  |            |       |          | 0.05402           | 0                          |

99. POU5F1B - hepatocellular carcinoma

ENSG00000212993 | EFO\_0000182

|                                     |                          |                                                          |
|-------------------------------------|--------------------------|----------------------------------------------------------|
| L2G score<br>0.828                  | Unmet need index<br>4.79 | SAS global MAF<br>0.001041                               |
| Genome India global MAF<br>0.002812 | Variants<br>2            | Interpretations<br>Likely consequential;<br>not_missense |

Disease Description

A malignant tumor that arises from hepatocytes. Hepatocellular carcinoma is relatively rare in the United States but very common in all African countries south of the Sahara and in Southeast Asia. Most cases are seen in patients over the age of 50 years, but this tumor can also occur in younger individuals and even in children. Hepatocellular carcinoma is more common in males than females and is associated with hepatitis B, hepatitis C, chronic alcohol abuse and cirrhosis. Serum elevation of alpha-fetoprotein occurs in a large percentage of patients with hepatocellular carcinoma. Grossly, hepatocellular carcinoma may present as a single mass, as multiple nodules, or as diffuse liver involvement. Microscopically, there is a wide range of differentiation from tumor to tumor (well differentiated to poorly differentiated tumors). Hepatocellular carcinomas quickly metastasize to regional lymph nodes and lung. The overall median survival of untreated liver cell carcinoma is about 4 months. The most effective treatment of hepatocellular carcinoma is complete resection of the tumor. Lately, an increasing number of tumors have been treated with liver transplantation.

Gene Summary

POU class 5 homeobox 1B

Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.828 across 3 evidence rows and 3 study loci. No datasource-level direct association summary was found in the downloaded association table.

SAS / Indian Prevalence

Rare

Alleles and Scores

| varid           | consequence      | missense_interpretation | AF_sas    | genome_india_af | AF_nfe | AF_fin | cadd_phred | am_pathogenicity | EVE_S CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|-----------------|------------------|-------------------------|-----------|-----------------|--------|--------|------------|------------------|------------|-------|----------|-------------------|----------------------------|
| 8_127416359_G_C | missense_variant | Likely consequential    | 0.0008054 | 0.001586        | 0      | 0      | 20.9       | 0.6828           |            |       | 0.9479   | 0.001041          | 0.002812                   |
| 8_127416785_G_T | stop_gained      | not_missense            | 0.0002362 | 0.001228        | 0      | 0      | 35         |                  |            |       |          | 0.001041          | 0.002812                   |

# 100. HNF1B - hepatocellular carcinoma

ENSG00000275410 | EFO\_0000182

|                                     |                                 |                                        |
|-------------------------------------|---------------------------------|----------------------------------------|
| L2G score<br><b>0.71</b>            | Unmet need index<br><b>4.79</b> | SAS global MAF<br><b>0.0169</b>        |
| Genome India global MAF<br><b>0</b> | Variants<br><b>1</b>            | Interpretations<br><b>not_missense</b> |

## Disease Description

A malignant tumor that arises from hepatocytes. Hepatocellular carcinoma is relatively rare in the United States but very common in all African countries south of the Sahara and in Southeast Asia. Most cases are seen in patients over the age of 50 years, but this tumor can also occur in younger individuals and even in children. Hepatocellular carcinoma is more common in males than females and is associated with hepatitis B, hepatitis C, chronic alcohol abuse and cirrhosis. Serum elevation of alpha-fetoprotein occurs in a large percentage of patients with hepatocellular carcinoma. Grossly, hepatocellular carcinoma may present as a single mass, as multiple nodules, or as diffuse liver involvement. Microscopically, there is a wide range of differentiation from tumor to tumor (well differentiated to poorly differentiated tumors). Hepatocellular carcinomas quickly metastasize to regional lymph nodes and lung. The overall median survival of untreated liver cell carcinoma is about 4 months. The most effective treatment of hepatocellular carcinoma is complete resection of the tumor. Lately, an increasing number of tumors have been treated with liver transplantation.

## Gene Summary

HNF1 homeobox B

## Known Drugs / Clinical Investigation

No same target-disease clinical indication found in the downloaded Open Targets drug tables.

## Biological Evidence

GWAS credible-set evidence supports this gene-disease pair with L2G score 0.71 across 2 evidence rows and 2 study loci. No datasource-level direct association summary was found in the downloaded association table.

## SAS / Indian Prevalence

Rare

## Alleles and Scores

| varid                  | consequence        | missense_interpretation | AF_sas | genome_india_af | AF_nfe | AF_fin  | cadd_phred | am_pat_hogenicity | EVE_S_CORE | REVEL | ClinPred | global_maf_af_sas | global_maf_genome_india_af |
|------------------------|--------------------|-------------------------|--------|-----------------|--------|---------|------------|-------------------|------------|-------|----------|-------------------|----------------------------|
| 17_37710582_GTCACCAT_G | frameshift_variant | not_missense            | 0.0169 |                 | 0.0023 | 0.02255 | 35         |                   |            |       |          | 0.0169            | 0                          |